
Lesson 2

Parametric Distributions

Reading: *Loss Models* Fifth Edition 4, 5.3—5.4

A **parametric distribution** is one that is defined by a fixed number of parameters. Examples of parametric distributions are the **exponential distribution** (parameter θ) and the **Pareto distribution** (parameters α, θ). Any distribution listed in the *Loss Models* appendix is parametric.

The alternative to a parametric distribution is a **data-dependent distribution**. A data-dependent distribution is one where the specification requires at least as many “parameters” as the number of data points in the sample used to create it; the bigger the sample, the more “parameters”.

It is common to use parametric distributions for short term insurance claim counts (**frequency**) and loss size (**severity**). Parametric distributions have many advantages. One of the advantages of parametric distributions which makes them so useful for severity is that they handle inflation easily.

2.1 Scaling

A parametric distribution is a member of a **scale family** if any positive multiple of the random variable has the same form. In other words, the distribution function of cX , for c a positive constant, is of the same form as the distribution function of X , but with different values for the parameters. Sometimes the distribution can be parametrized in such a way that only one parameter of cX has a value different from the parameters of X . If the distribution is parametrized in this fashion, so that the only parameter of cX having a different value from X is θ , and the value of θ for cX is c times the value of θ for X , then θ is called a **scale parameter**.

All of the continuous distributions in the tables (Appendix A) are scale families. The parametrizations given in the tables are often different from those you would find in other sources, such as your probability textbook. They are parametrized so that θ is the scale parameter. Thus when you are given that a random variable has any distribution in the appendix and you are given the parameters, it is easy to determine the distribution of a multiple of the random variable.

The only distributions not parametrized with a scale parameter are the **lognormal** and the **inverse Gaussian**. Even though the inverse Gaussian has θ as a parameter, it is not a scale parameter. The parametrization for the lognormal given in the tables is the traditional one. *If you need to scale a lognormal, proceed as follows: if X is lognormal with parameters (μ, σ) , then cX is lognormal with parameters $(\mu + \ln c, \sigma)$.*

To scale a random variable not in the tables, you’d reason as follows. Let $Y = cX$, $c > 0$. Then

$$F_Y(y) = \Pr(Y \leq y) = \Pr(cX \leq y) = \Pr\left(X \leq \frac{y}{c}\right) = F_X\left(\frac{y}{c}\right)$$

One use of scaling is in handling inflation. In fact, handling inflation is the only topic in this lesson that is commonly tested directly. If loss sizes are inflated by $100r\%$, the **inflated loss variable** Y will be $(1 + r)X$, where X is the pre-inflation loss variable. For a scale family with a scale parameter, you just multiply θ by $(1 + r)$ to obtain the new distribution.

EXAMPLE 2A Claim sizes expressed in dollars follow a two-parameter Pareto distribution with parameters $\alpha = 5$ and $\theta = 90$. A euro is worth \$1.50.

Calculate the probability that a claim will be for 20 euros or less.

SOLUTION: If claim sizes in dollars are X , then claim sizes in euros are $Y = X/1.5$. The resulting euro-based random variable Y for claim size will be Pareto with $\alpha = 5$, $\theta = 90/1.5 = 60$. The probability that a claim will be no more

than 20 euros is

$$\Pr(Y \leq 20) = F_Y(20) = 1 - \left(\frac{60}{60 + 20}\right)^5 = \boxed{0.7627}$$

□

EXAMPLE 2B Claim sizes in 2022 follow a lognormal distribution with parameters $\mu = 4.5$ and $\sigma = 2$. Claim sizes grow at 6% uniform inflation during 2023 and 2024.

Calculate $f(1000)$, the probability density function at 1000, of the claim size distribution in 2024.

SOLUTION: If X is the claim size random variable in 2022, then $Y = 1.06^2 X$ is the revised variable in 2024. The revised lognormal distribution of Y has parameters $\mu = 4.5 + 2 \ln 1.06$ and $\sigma = 2$. The probability density function at 1000 is

$$\begin{aligned} f_Y(1000) &= \frac{1}{\sigma(1000)\sqrt{2\pi}} e^{-(\ln 1000 - \mu)^2 / 2\sigma^2} \\ &= \frac{1}{(2)(1000)\sqrt{2\pi}} e^{-[\ln 1000 - (4.5 + 2 \ln 1.06)]^2 / 2(2^2)} \\ &= (0.000199471)(0.518814) = \boxed{0.0001035} \end{aligned}$$

□

EXAMPLE 2C Claim sizes expressed in dollars follow a lognormal distribution with parameters $\mu = 3$ and $\sigma = 2$. A euro is worth \$1.50.

Calculate the probability that a claim will be for 100 euros or less.

SOLUTION: If claim sizes in dollars are X , then claim sizes in euros are $Y = X/1.5$. As discussed above, the distribution of claim sizes in euros is lognormal with parameters $\mu = 3 - \ln 1.5$ and $\sigma = 2$. Then

$$F_Y(100) = \Phi\left(\frac{\ln 100 - 3 + \ln 1.5}{2}\right) = \Phi(1.01) = \boxed{0.8438}$$

□

EXAMPLE 2D Claim sizes X initially follow a distribution with distribution function:

$$F_X(x) = 1 - \frac{1}{e^{0.01x}(1 + 0.01x)} \quad x > 0$$

Claim sizes are inflated by 50% uniformly.

Calculate the probability that a claim will be for 60 or less after inflation.

SOLUTION: Let Y be the increased claim size. Then $Y = 1.5X$, so $\Pr(Y \leq 60) = \Pr(X \leq 60/1.5) = F_X(40)$.

$$F_X(40) = 1 - \frac{1}{1.4e^{0.4}} = \boxed{0.5212}$$

□

2.2 Common parametric distributions

The tables provide a lot of information about the distributions, but if you don't recognize the distribution, you won't know to use the table. Therefore, it is a good idea to be familiar with the common distributions.

You should familiarize yourself with the *form* of each distribution, but not necessarily the constants. The constant is forced so that the density function will integrate to 1. If you know which distribution you are dealing with, you can figure out the constant. To emphasize this point, in the following discussion, we will use the letter c for constants rather than spelling out what the constants are. You are not trying to recognize the constant; you are trying to recognize the form.

We will mention the means and variances or second moments of the distributions. You need not memorize any of these. The tables give you the **raw moments**. You can calculate the **variance** as $E[X^2] - E[X]^2$. However, for

The gamma function

The **gamma function** $\Gamma(x)$ is a generalization to real numbers of the factorial function, defined by

$$\Gamma(x) = \int_0^{\infty} u^{x-1} e^{-u} du$$

For positive integers n ,

$$\Gamma(n) = (n-1)!$$

The most important relationship for $\Gamma(x)$ that you should know is

$$\Gamma(x+1) = x\Gamma(x)$$

for any real number x .

EXAMPLE 2E Evaluate $\frac{\Gamma(8.5)}{\Gamma(6.5)}$.

SOLUTION:

$$\frac{\Gamma(8.5)}{\Gamma(6.5)} = \left(\frac{\Gamma(8.5)}{\Gamma(7.5)} \right) \left(\frac{\Gamma(7.5)}{\Gamma(6.5)} \right) = (7.5)(6.5) = \boxed{48.75}$$

frequently used distributions, you may want to memorize the mean and variance to save yourself some time when working out questions.

We will graph the distributions. You are not responsible for graphs, but they may help you understand the distributions.

The tables occasionally use the **gamma function** $\Gamma(x)$ in the formulas for the moments. You should have a basic knowledge of the gamma function; if you are not familiar with this function, see the sidebar. The tables also define and use the **incomplete gamma** and **incomplete beta** functions, but you can get by without knowing them.¹

2.2.1 Uniform

A **uniform distribution** has a constant density on $[d, u]$:

$$f(x; d, u) = \frac{1}{u-d} \quad d \leq x \leq u$$

$$F(x; d, u) = \begin{cases} 0 & x \leq d \\ \frac{x-d}{u-d} & d \leq x \leq u \\ 1 & x \geq u \end{cases}$$

You recognize a uniform distribution both by its finite **support**² and by the lack of an x in the density function.

Its moments are

$$E[X] = \frac{d+u}{2}$$

$$\text{Var}(X) = \frac{(u-d)^2}{12}$$

¹But Excel provides these functions. If you are taking Exam ASTAM, an exam that provides you with a spreadsheet, you may need to use these functions.

²"Support" is the range of values for which the probability density function is nonzero.

Its mean, median, and midrange are equal. The best way to calculate the second moment is to add up the variance and the square of the mean. However, some students prefer to use the following easy-to-derive formula:

$$\mathbf{E}[X^2] = \frac{1}{u-d} \int_d^u x^2 dx = \frac{u^3 - d^3}{3(u-d)} = \frac{u^2 + ud + d^2}{3} \quad (2.1)$$

If $d = 0$, then the formula reduces to $u^2/3$.

The uniform distribution is not directly in the tables, so I recommend you memorize the formulas for mean and variance. However, if $d = 0$, then the uniform distribution is a special case of a beta distribution with $\theta = u$, $a = 1$, $b = 1$.

2.2.2 Beta



The probability density function of a **beta distribution** with $\theta = 1$ has the form

$$f(x; a, b) = cx^{a-1}(1-x)^{b-1} \quad 0 \leq x \leq 1$$

The parameters a and b must be positive. They may equal 1, in which case the corresponding factor is missing from the density function. Thus if $a = b = 1$, the beta distribution is a uniform distribution.

You recognize a beta distribution both by its finite support—it's the only common distribution for which the density is nonzero only on a finite range of values—and by factors with x and $1 - x$ raised to powers and no other use of x in the density function.

If θ is arbitrary, then the form of the probability density function is

$$f(x; a, b, \theta) = cx^{a-1}(\theta - x)^{b-1} \quad 0 \leq x \leq \theta$$

The distribution function can be evaluated if a or b is an integer. The moments are

$$\begin{aligned} \mathbf{E}[X] &= \frac{\theta a}{a+b} \\ \text{Var}(X) &= \frac{\theta^2 ab}{(a+b)^2(a+b+1)} \end{aligned}$$

The mode is $\theta(a-1)/(a+b-2)$ when a and b are both greater than 1, but you are not responsible for this fact.

Figure 2.1 graphs four beta distributions with $\theta = 1$ all having mean $2/3$. You can see how the distribution becomes more peaked and normal looking as a and b increase.

2.2.3 Exponential



The probability density function of an **exponential distribution** has the form

$$f(x; \theta) = ce^{-x/\theta} \quad x \geq 0$$

θ must be positive.

You recognize an exponential distribution when the density function has e raised to a multiple of x , and no other use of x .

The distribution function is easily evaluated. The moments are:

$$\begin{aligned} \mathbf{E}[X] &= \theta \\ \text{Var}(X) &= \theta^2 \end{aligned}$$

Figure 2.2 graphs three exponential distributions. The higher the parameter, the more weight placed on higher numbers.

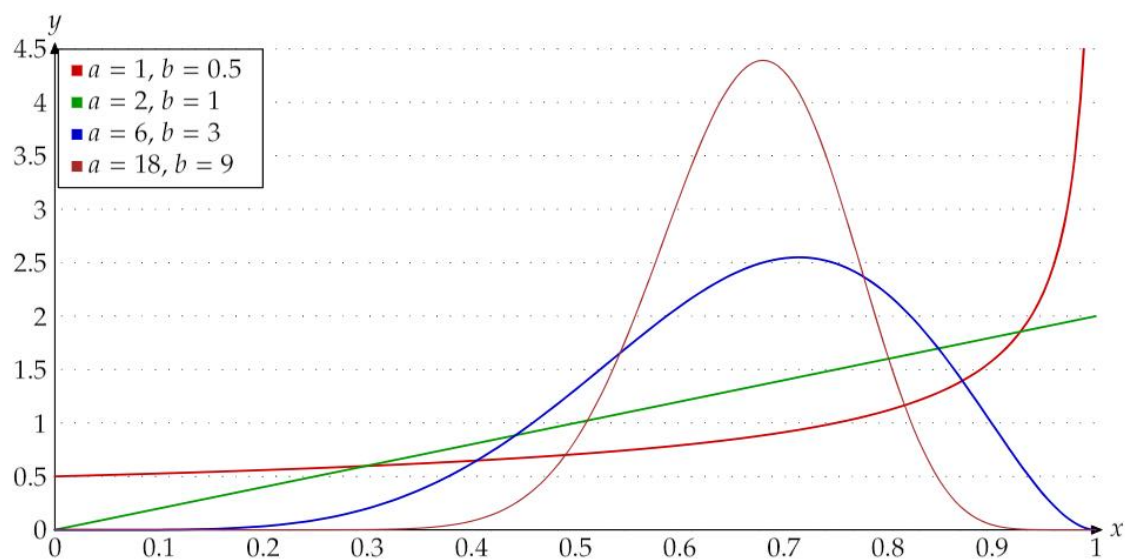


Figure 2.1: Probability density function of four beta distributions with $\theta = 1$ and mean $2/3$

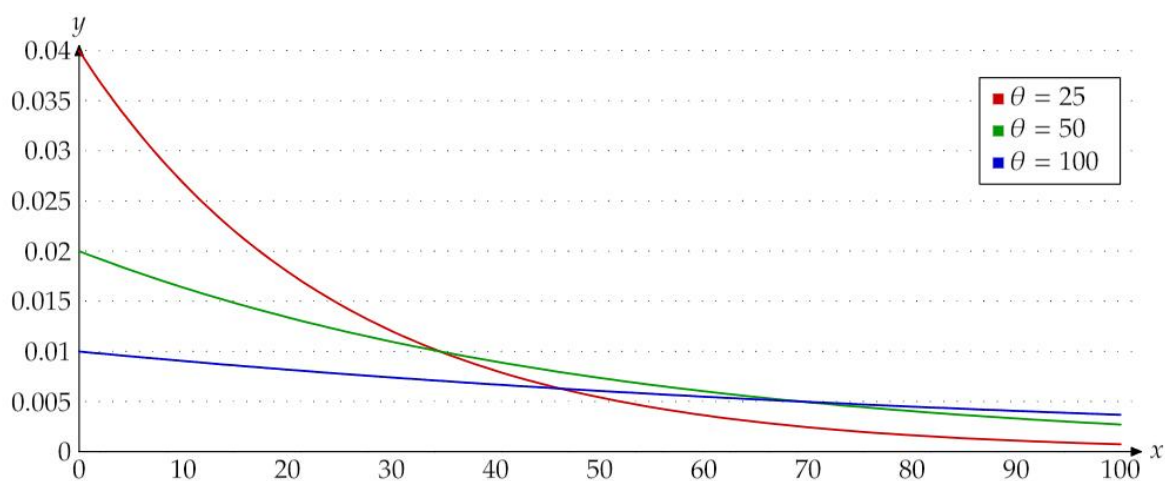


Figure 2.2: Probability density function of three exponential distributions

2.2.4 Weibull

A **Weibull distribution** is a transformed exponential distribution. If Y is exponential with mean μ , then $X = Y^{1/\tau}$ is Weibull with parameters $\theta = \mu^{1/\tau}$ and τ . An exponential is a special case of a Weibull with $\tau = 1$.

The form of the density function is

$$f(x; \tau, \theta) = c x^{\tau-1} e^{-(x/\theta)^\tau} \quad x \geq 0$$

Both parameters must be positive.

You recognize a Weibull distribution when the density function has e raised to a multiple of a power of x , and in addition has a corresponding power of x , one lower than the power in the exponential, as a factor.

The distribution function is easily evaluated, but the moments require evaluating the **gamma function**, which usually requires numerical techniques. The moments are

$$E[X] = \theta \Gamma(1 + 1/\tau)$$

$$E[X^2] = \theta^2 \Gamma(1 + 2/\tau)$$

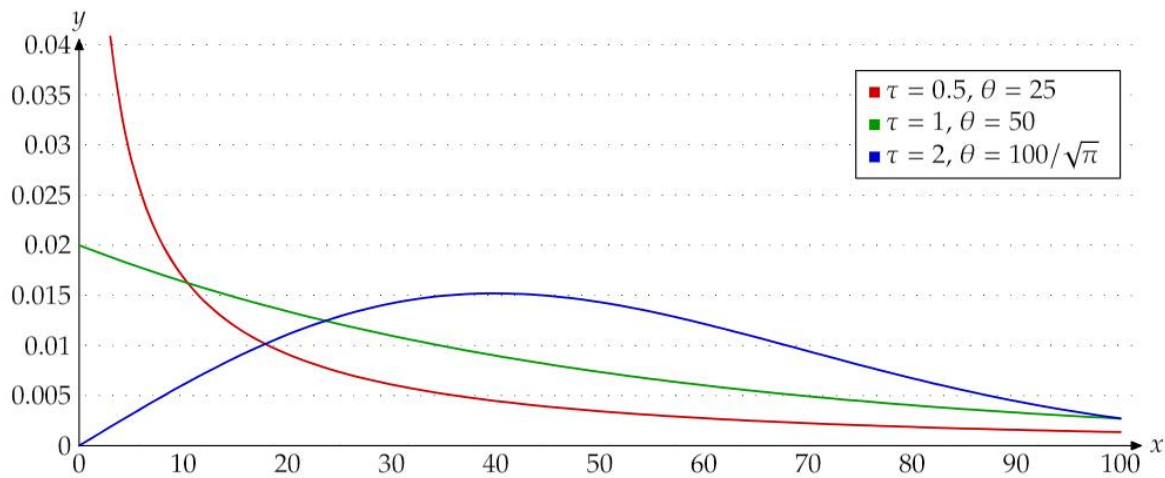


Figure 2.3: Probability density function of three Weibull distributions with mean 50

Figure 2.3 graphs three Weibull distributions with mean 50. The distribution has a non-zero mode when $\tau > 1$. Notice that the distribution with $\tau = 0.5$ puts a lot of weight on small numbers. To make up for this, it will also have to put higher weight than the other two distributions on very large numbers, so although it's not shown, its graph will cross the other two graphs for high x .

2.2.5 Gamma

The form of the density function of a **gamma distribution** is

$$f(x; \alpha, \theta) = cx^{\alpha-1}e^{-x/\theta} \quad x \geq 0$$

Both parameters must be positive.

When α is an integer, a gamma random variable with parameters α and θ is the sum of α independent exponential random variables with parameter θ . In particular, when $\alpha = 1$, the gamma random variable is exponential. The gamma distribution is called an **Erlang distribution** when α is an integer.

You recognize a gamma distribution when the density function has e raised to a multiple of x , and in addition has x raised to a power. Contrast this with a Weibull, where e is raised to a multiple of a *power* of x .

The distribution function may be evaluated if α is an integer; otherwise numerical techniques are needed. However, the moments are easily evaluated:

$$\begin{aligned} E[X] &= \alpha\theta \\ \text{Var}(X) &= \alpha\theta^2 \end{aligned}$$

Figure 2.4 graphs three gamma distributions with mean 50. As α goes to infinity, the graph's peak narrows and the distribution converges to a normal distribution.

The gamma distribution is one of the few for which the moment generating function has a closed form. In particular, the moment generating function of an exponential has a closed form. The only other distributions in the tables with closed form moment generating functions are the normal distribution (not actually in the tables, but the formula for the lognormal moments is the MGF of a normal) and the inverse Gaussian.

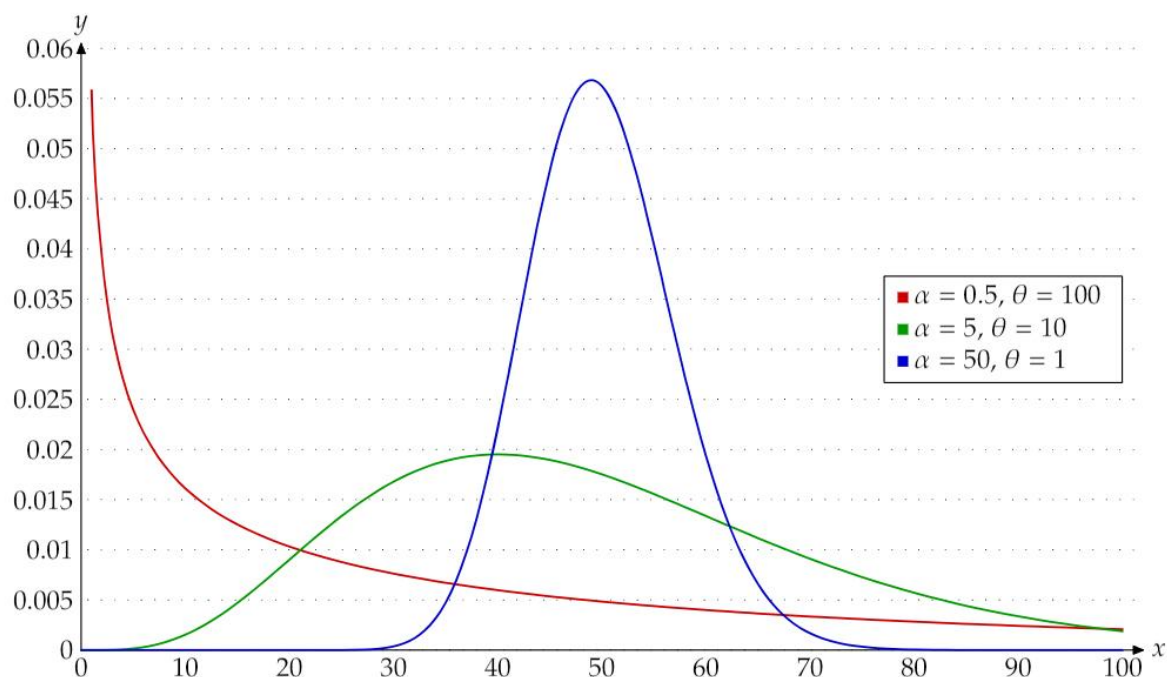


Figure 2.4: Probability density function of three gamma distributions with mean 50

2.2.6 Pareto

When we say “Pareto”, we mean a *two-parameter Pareto*. On recent exams, they write out “two-parameter” to make it clear, but on older exams, you will often find the word “Pareto” with no qualifier. It always refers to a two-parameter Pareto, not a single-parameter Pareto.

The form of the density function of a **two-parameter Pareto** is

$$f(x) = \frac{c}{(\theta + x)^{c+1}} \quad x \geq 0$$

Both parameters must be positive.

You recognize a Pareto when the density function has a denominator with x plus a constant raised to a power.

The distribution function is easily evaluated. The moments are

$$\begin{aligned} E[X] &= \frac{\theta}{\alpha - 1} & \alpha > 1 \\ E[X^2] &= \frac{2\theta^2}{(\alpha - 1)(\alpha - 2)} & \alpha > 2 \end{aligned}$$

When α does not satisfy these conditions, the corresponding moments don't exist.

A shortcut formula for the variance of a Pareto is

$$\text{Var}(X) = E[X]^2 \left(\frac{\alpha}{\alpha - 2} \right)$$

Figure 2.5 graphs three Pareto distributions, one with $\alpha < 1$ and the other two with mean 50. Although the one with $\alpha = 0.5$ puts higher weight on small numbers than the other two, its mean is infinite; it puts higher weight on large numbers than the other two, and its graph eventually crosses the other two as $x \rightarrow \infty$.

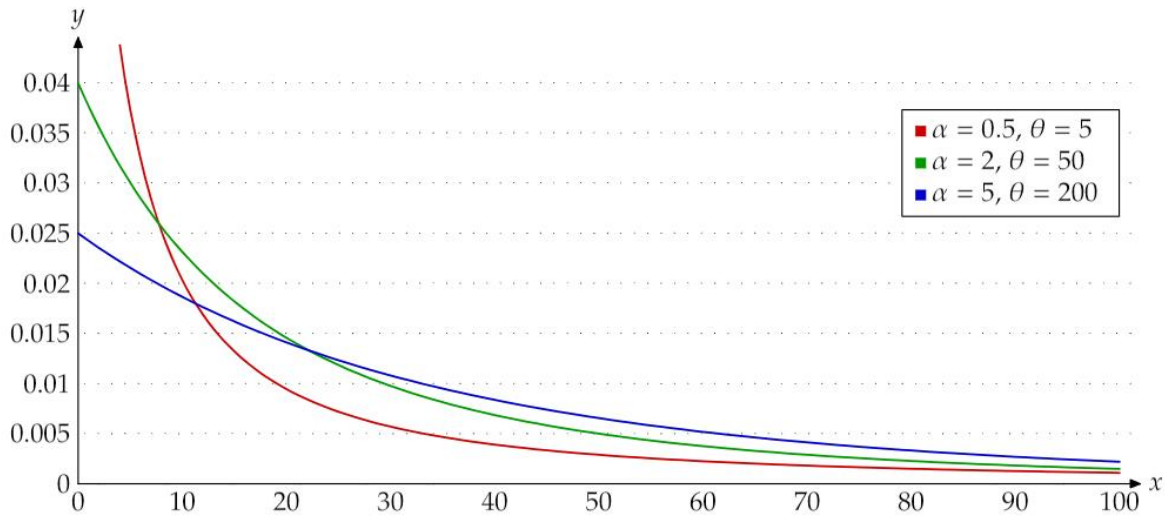


Figure 2.5: Probability density function of three Pareto distributions

2.2.7 Single-parameter Pareto

The form of the density function of a **single-parameter Pareto** is

$$f(x) = \frac{c}{x^{\alpha+1}} \quad x \geq \theta$$

α must be positive. θ is not considered a parameter since it must be selected in advance, based on what you want the range to be.

You recognize a single-parameter Pareto by the range of nonzero values for its density function—unlike most other distributions, this range does not start at 0—and by the form of the density function, which has a denominator with x raised to a power. A beta distribution may also have x raised to a negative power, but its density function is 0 above a finite number.

A single-parameter Pareto X is a two-parameter Pareto Y shifted by θ : $X = Y + \theta$. Thus it has the same variance, and the mean is θ greater than the mean of a two-parameter Pareto with the same parameters.

$$E[X] = \frac{\alpha\theta}{\alpha - 1} \quad \alpha > 1$$

$$E[X^2] = \frac{\alpha\theta^2}{\alpha - 2} \quad \alpha > 2$$

2.2.8 Lognormal

The form of the density function of a **lognormal distribution** is

$$f(x) = \frac{ce^{-(\ln x - \mu)^2/2\sigma^2}}{x} \quad x > 0$$

σ must be nonnegative.

You recognize a lognormal by the $\ln x$ in the exponent.

If Y is normal, then $X = e^Y$ is lognormal with the same parameters μ and σ . Thus, to calculate the distribution function, use

$$F_X(x) = F_Y(\ln x) = \Phi\left(\frac{\ln x - \mu}{\sigma}\right)$$

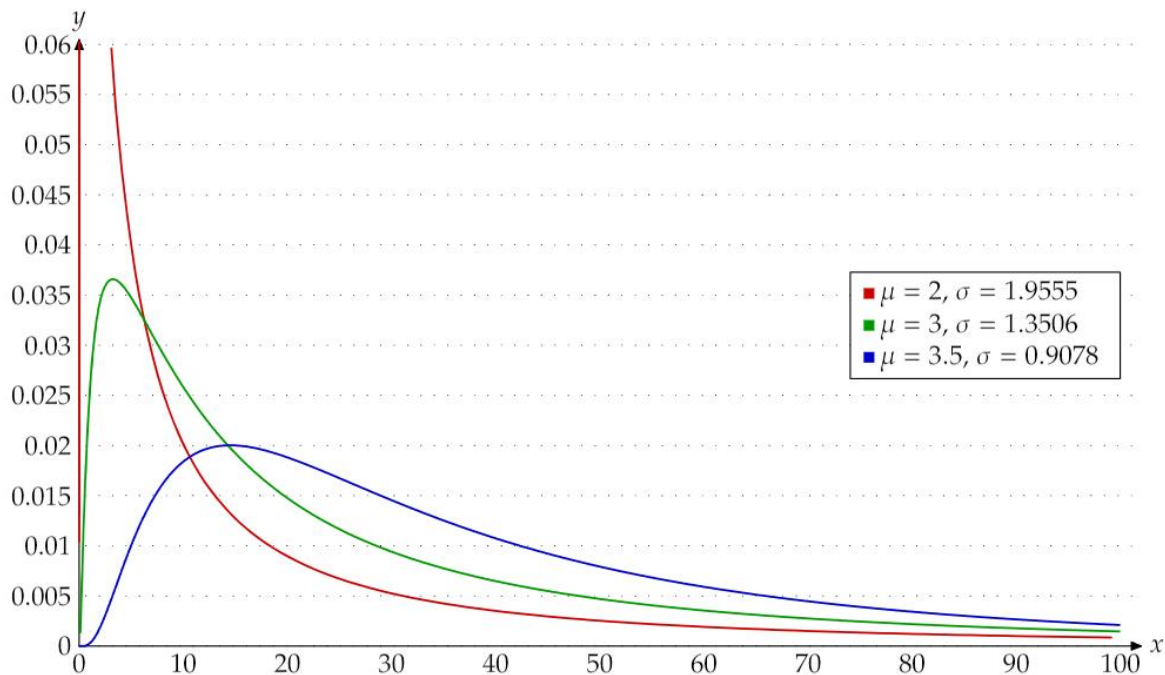


Figure 2.6: Probability density function of three lognormal distributions with mean 50

where $\Phi(x)$ is the standard normal distribution function, for which you are given tables. The moments of a lognormal are

$$\begin{aligned} \mathbf{E}[X] &= e^{\mu+0.5\sigma^2} \\ \mathbf{E}[X^2] &= e^{2\mu+2\sigma^2} \end{aligned}$$

More generally, $\mathbf{E}[X^k] = \mathbf{E}[e^{kY}] = M_Y(k)$, where $M_Y(k)$ is the moment generating function of the corresponding normal distribution.

Figure 2.6 graphs three lognormals with mean 50. The mode is $\exp(\mu - \sigma^2)$, as stated in the tables. For $\mu = 2$, the mode is off the graph. As σ gets lower, the distribution flattens out.

Table 2.1 is a summary of the forms of probability density functions for common distributions.

2.3 The linear exponential family

The following material is based on *Loss Models* 5.4 which is on the syllabus. It won't play any role in this course, but the **linear exponential family** is commonly used for generalized linear models³, which you'll study when working on Exam SRM. I doubt anything in this section will be tested on directly, so you may skip it.

A set of parametric distributions is in the linear exponential family if it can be parametrized with a parameter θ in such a way that in its density function, the only interaction between θ and x is in the exponent of e , which is x times a function of θ . In other words, its density function $f(x; \theta)$ can be expressed as

$$f(x; \theta) = \frac{p(x)e^{r(\theta)x}}{q(\theta)}$$

The set may have other parameters. $q(\theta)$ is the normalizing constant which makes the integral of f equal to 1. $r(\theta)$ is called the **canonical parameter** of the distribution.

³Many textbooks leave out "linear" and just call it the exponential family.

Table 2.1: Forms of probability density functions for common distributions

Distribution	Probability density function	
Uniform	c	$d \leq x \leq u$
Beta	$cx^{a-1}(\theta - x)^{b-1}$	$0 \leq x \leq \theta$
Exponential	$ce^{-x/\theta}$	$x \geq 0$
Weibull	$cx^{\tau-1}e^{-x^\tau/\theta^\tau}$	$x \geq 0$
Gamma	$cx^{\alpha-1}e^{-x/\theta}$	$x \geq 0$
Pareto	$\frac{c}{(x + \theta)^{\alpha+1}}$	$x \geq 0$
Single-parameter Pareto	$\frac{c}{x^{\alpha+1}}$	$x \geq \theta$
Lognormal	$\frac{ce^{-(\ln x - \mu)^2/2\sigma^2}}{x}$	$x > 0$

Examples of the linear exponential family are:

Gamma distribution The pdf is

$$f(x; \mu, \sigma) = \frac{x^{\alpha-1}e^{-x/\theta}}{\Gamma(\alpha)\theta^\alpha}$$

Let $r(\theta) = -1/\theta$, $p(x) = x^{\alpha-1}$, and $q(\theta) = \Gamma(\alpha)\theta^\alpha$.

Normal distribution The pdf is

$$f(x; \theta) = \frac{e^{-(x-\mu)^2/2\sigma^2}}{\sigma\sqrt{2\pi}}$$

Let $\theta = \mu$. The denominator of the pdf does not have x or θ so it can go into $q(\theta)$ or into $p(x)$. The exponent can be expanded into

$$-\frac{x^2}{2\sigma^2} + \frac{x\theta}{\sigma^2} - \frac{\theta^2}{2\sigma^2}$$

and only the second summand involves both x and θ , and x appears to the first power. Thus we can set $p(x) = e^{-x^2/2\sigma^2}$, $r(\theta) = \theta/\sigma^2$, and $q(\theta) = e^{\theta^2/2\sigma^2}\sigma\sqrt{2\pi}$.

Discrete distributions are in the linear exponential family if we can express the probability function in the linear exponential form.

Poisson distribution For a Poisson distribution, the probability function is

$$f(x; \lambda) = e^{-\lambda} \frac{\lambda^x}{x!} = e^{-\lambda} \frac{e^{x \ln \lambda}}{x!}$$

We can let $\theta = \lambda$, and then $p(x) = 1/x!$, $r(\theta) = \ln \theta$, and $q(\theta) = e^\theta$.

The textbook develops the following formulas for the mean and variance of a distribution from the linear exponential family:

$$\begin{aligned} \mathbf{E}[X] = \mu(\theta) &= \frac{q'(\theta)}{r'(\theta)q(\theta)} = \frac{(\ln q(\theta))'}{r'(\theta)} \\ \text{Var}(X) = v(\theta) &= \frac{\mu'(\theta)}{r'(\theta)} \end{aligned}$$

Thus, in the above examples:

Gamma distribution

$$\begin{aligned} \frac{d \ln q}{d\theta} &= \frac{\alpha}{\theta} \\ \frac{dr}{d\theta} &= \frac{1}{\theta^2} \\ \mathbf{E}[X] &= \frac{\alpha/\theta}{1/\theta^2} = \alpha\theta \\ \text{Var}(X) &= \frac{\alpha}{1/\theta^2} = \alpha\theta^2 \end{aligned}$$

Normal distribution

$$\begin{aligned} (\ln q(\theta))' &= \frac{2\theta}{2\sigma^2} = \frac{\theta}{\sigma^2} \\ r'(\theta) &= \frac{1}{\sigma^2} \\ \mathbf{E}[X] &= \frac{\theta/\sigma^2}{1/\sigma^2} = \theta \\ \text{Var}(X) &= \frac{1}{1/\sigma^2} = \sigma^2 \end{aligned}$$

Poisson distribution

$$\begin{aligned} (\ln q(\theta))' &= 1 \\ r'(\theta) &= \frac{1}{\theta} \\ \mathbf{E}[X] &= \frac{1}{1/\theta} = \theta \\ \text{Var}(X) &= \frac{1}{1/\theta} = \theta \end{aligned}$$

2.4 Limiting distributions

The following material is based on *Loss Models* 5.3.3. I don't think it has ever appeared on the exam and doubt it ever will.

In some cases, as the parameters of a distribution go to infinity, the distribution converges to another distribution. To demonstrate this, we will usually have to use the identity

$$\lim_{\alpha \rightarrow \infty} \left(1 + \frac{r}{\alpha}\right)^\alpha = e^r$$

Equivalently, if c is a constant (not dependent on α), then

$$\lim_{\alpha \rightarrow \infty} \left(1 + \frac{r}{\alpha}\right)^{\alpha+c} = \lim_{\alpha \rightarrow \infty} \left(1 + \frac{r}{\alpha}\right)^{\alpha} \left(1 + \frac{r}{\alpha}\right)^c = e^r$$

As a simple example (not in the textbook) of a limiting distribution, consider a **gamma distribution** with a fixed mean μ , and let $\alpha \rightarrow \infty$. Then $\theta = \mu/\alpha$. The **moment generating function** is

$$M(t) = (1 - \theta t)^{-\alpha} = \frac{1}{\left(1 - \frac{\mu t}{\alpha}\right)^{\alpha}}$$

and as $\alpha \rightarrow \infty$, the denominator goes to $e^{-\mu t}$, so $M(t) \rightarrow e^{\mu t}$, which is the moment generating function of the constant μ . So as $\alpha \rightarrow \infty$, the limiting distribution of a gamma is a distribution equal to the mean with probability 1.

As another example, let's carry out textbook exercise 5.21, which asks you to demonstrate that the limiting distribution of a **Pareto** with θ/α constant as $\alpha \rightarrow \infty$ is an exponential. Let $k = \theta/\alpha$. The density function of a Pareto is

$$\begin{aligned} f(x; \alpha, \theta) &= \frac{\alpha \theta^{\alpha}}{(\theta + x)^{\alpha+1}} = \frac{\alpha (\alpha k)^{\alpha}}{(\alpha k + x)^{\alpha+1}} \\ &= \frac{k^{\alpha}}{(k + x/\alpha)^{\alpha+1}} = \frac{1}{k(1 + (x/k)/\alpha)^{\alpha+1}} \end{aligned}$$

and the limit as $\alpha \rightarrow \infty$ is $(1/k)e^{-x/k}$. That is the density function of an exponential with mean k . Notice that as $\alpha \rightarrow \infty$, the mean of the Pareto converges to k .

Table 2.2: Summary of Parametric Distribution Concepts

- If X is a member of a **scale family** with **scale parameter** θ with value s , then cX is in the same family and has the same parameter values as X except that the scale parameter θ has value cs .
- All distributions in the tables are scale families with scale parameter θ except for **lognormal** and **inverse Gaussian**.
- If X is lognormal with parameters μ and σ , then cX is lognormal with parameters $\mu + \ln c$ and σ .
- If X is normal with parameters μ and σ^2 , then e^X is lognormal with parameters μ and σ .
- See Table 2.1 to learn the forms of commonly occurring distributions. Useful facts are

Uniform on $[d, u]$	$E[X] = \frac{d+u}{2}$
	$\text{Var}(X) = \frac{(u-d)^2}{12}$

Uniform on $[0, u]$	$E[X^2] = \frac{u^2}{3}$
Gamma	$\text{Var}(X) = \alpha \theta^2$

- If Y is **single-parameter Pareto** with parameters α and θ , then $Y - \theta$ is **two-parameter Pareto** with the same parameters.
- X is in the **linear exponential family** if its probability density function can be expressed as

$$f(x; \theta) = \frac{p(x)e^{r(\theta)x}}{q(\theta)}$$

Exercises

- 2.1.  Loss sizes for an insurance coverage follow an inverse gamma distribution with mean 6 and mode 4. Calculate the coefficient of skewness for the losses.
- (A) 3.1 (B) 3.2 (C) 3.3 (D) 3.4 (E) 3.5
- 2.2.  For a commercial fire coverage
- In 2023, loss sizes follow a two-parameter Pareto distribution with parameters $\alpha = 4$ and θ .
 - In 2024, there is uniform inflation at rate r .
 - The 65th percentile of loss size in 2024 equals the mean loss size in 2023.
- Determine r .
- 2.3.  [CAS3-S06:26] The aggregate losses of Eiffel Auto Insurance are denoted in euro currency and follow a lognormal distribution with $\mu = 8$ and $\sigma = 2$.
- Given that 1 euro = 1.3 dollars, which set of lognormal parameters describes the distribution of Eiffel's losses in dollars?
- (A) $\mu = 6.15, \sigma = 2.26$
(B) $\mu = 7.74, \sigma = 2.00$
(C) $\mu = 8.00, \sigma = 2.60$
(D) $\mu = 8.26, \sigma = 2.00$
(E) $\mu = 10.40, \sigma = 2.60$
- 2.4.  [4B-S90:37] (2 points) Liability claim severity follows a Pareto distribution with a mean of 25,000 and parameter $\alpha = 3$.
- If inflation increases all claims by 20%, the probability of a claim exceeding 100,000 increases by what amount?
- (A) Less than 0.02
(B) At least 0.02, but less than 0.03
(C) At least 0.03, but less than 0.04
(D) At least 0.04, but less than 0.05
(E) At least 0.05

2.5.  [4B-F97:26] (3 points) You are given the following:

- In 1996, losses follow a lognormal distribution with parameters μ and σ .
- In 1997, losses follow a lognormal distribution with parameters $\mu + \ln k$ and σ , where k is greater than 1.
- In 1996, 100 p % of the losses exceed the mean of the losses in 1997.

Determine σ .

Note: z_p is the 100 p th percentile of a normal distribution with mean 0 and variance 1.

- (A) $2 \ln k$
- (B) $-z_p \pm \sqrt{z_p^2 - 2 \ln k}$
- (C) $z_p \pm \sqrt{z_p^2 - 2 \ln k}$
- (D) $\sqrt{-z_p \pm \sqrt{z_p^2 - 2 \ln k}}$
- (E) $\sqrt{z_p \pm \sqrt{z_p^2 - 2 \ln k}}$

2.6.  [4B-S94:16] (1 point) You are given the following:

- Losses in 1993 follow the density function

$$f(x) = 3x^{-4}, \quad x \geq 1,$$

where x = losses in millions of dollars.

- Inflation of 10% impacts all claims uniformly from 1993 to 1994.

Determine the probability that losses in 1994 exceed 2.2 million.

- (A) Less than 0.05
- (B) At least 0.05, but less than 0.10
- (C) At least 0.10, but less than 0.15
- (D) At least 0.15, but less than 0.20
- (E) At least 0.20

2.7.  [4B-F95:6] (2 points) You are given the following:

- In 1994, losses follow a Pareto distribution with parameters $\theta = 500$ and $\alpha = 1.5$.
- Inflation of 5% impacts all losses uniformly from 1994 to 1995.

What is the median of the portion of the 1995 loss distribution above 200?

- (A) Less than 600
- (B) At least 600, but less than 620
- (C) At least 620, but less than 640
- (D) At least 640, but less than 660
- (E) At least 660

- 2.8. [CAS3-S04:34] Claim severities are modeled using a continuous distribution and inflation impacts claims uniformly at an annual rate of i .

Which of the following are true statements regarding the distribution of claim severities after the effect of inflation?

1. An Exponential distribution will have scale parameter $(1 + i)\theta$
2. A 2-parameter Pareto distribution will have scale parameters $(1 + i)\alpha$ and $(1 + i)\theta$.
3. A Paralogistic distribution will have scale parameter $\theta/(1 + i)$

(A) 1 only (B) 3 only (C) 1 and 2 only (D) 2 and 3 only (E) 1, 2, and 3

- 2.9. [CAS3-F05:21] Losses during the current year follow a Pareto distribution with $\alpha = 2$ and $\theta = 400,000$. Annual inflation is 10%.

Calculate the ratio of the expected proportion of claims that will exceed \$750,000 next year to the proportion of claims that exceed \$750,000 this year.

- (A) Less than 1.105
(B) At least 1.105, but less than 1.115
(C) At least 1.115, but less than 1.125
(D) At least 1.125, but less than 1.135
(E) At least 1.135

- 2.10. [4B-S99:17] You are given the following:

- In 1998, claim sizes follow a Pareto distribution with parameters θ (unknown) and $\alpha = 2$.
- Inflation of 6% affects all claims uniformly from 1998 to 1999.
- r is the ratio of the proportion of claims that exceed d in 1999 to the proportion of claims that exceed d in 1998.

Determine the limit of r as d goes to infinity.

- (A) Less than 1.05
(B) At least 1.05, but less than 1.10
(C) At least 1.10, but less than 1.15
(D) At least 1.15, but less than 1.20
(E) At least 1.20

2.11. [4B-F94:28] (2 points) You are given the following:

- In 1993, the claim amounts for a certain line of business were normally distributed with mean $\mu = 1000$ and variance $\sigma^2 = 10,000$;

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right) \quad -\infty < x < \infty, \quad \mu = 1000, \sigma = 100.$$

- Inflation of 5% impacted all claims uniformly from 1993 to 1994.

What is the distribution for claim amounts in 1994?

- (A) No longer a normal distribution
- (B) Normal with $\mu = 1000$ and $\sigma = 102.5$.
- (C) Normal with $\mu = 1000$ and $\sigma = 105.0$.
- (D) Normal with $\mu = 1050$ and $\sigma = 102.5$.
- (E) Normal with $\mu = 1050$ and $\sigma = 105.0$.

2.12. [4B-S93:11] (1 point) You are given the following:

- i) The underlying distribution for 1992 losses is given by a lognormal distribution with parameters $\mu = 17.953$ and $\sigma = 1.6028$.
- ii) Inflation of 10% impacts all claims uniformly the next year.

What is the underlying loss distribution after one year of inflation?

- (A) Lognormal with $\mu' = 19.748$ and $\sigma' = 1.6028$.
- (B) Lognormal with $\mu' = 18.048$ and $\sigma' = 1.6028$.
- (C) Lognormal with $\mu' = 17.953$ and $\sigma' = 1.7631$.
- (D) Lognormal with $\mu' = 17.953$ and $\sigma' = 1.4571$.
- (E) No longer a lognormal distribution

2.13. X follows an exponential distribution with mean 10.

Determine the mean of X^4 .

2.14. You are given

- i) X is exponential with mean 2.
- ii) $Y = X^{1.5}$.

Calculate $E[Y^2]$.

2.15. X follows a gamma distribution with parameters $\alpha = 2.5$ and $\theta = 10$.

$Y = 1/X$.

Evaluate $\text{Var}(Y)$.

Solutions

2.1. Looking up the tables for the inverse gamma distribution, we see that the mode is $\frac{\theta}{\alpha+1}$ and the mean is $\frac{\theta}{\alpha-1}$, so

$$\begin{aligned} \frac{\theta}{\alpha+1} &= 4 \\ \frac{\theta}{\alpha-1} &= 6 \end{aligned}$$

Dividing the second line into the first,

$$\frac{\alpha - 1}{\alpha + 1} = \frac{4}{6}$$

$$\alpha = 5 \quad \theta = 24$$

Then (γ_1 is the coefficient of skewness).

$$\begin{aligned} \mathbf{E}[X^2] &= \frac{\theta^2}{(\alpha - 1)(\alpha - 2)} = \frac{24^2}{12} = 48 \\ \text{Var}(X) &= 48 - 6^2 = 12 \\ \mathbf{E}[X^3] &= \frac{\theta^3}{(\alpha - 1)(\alpha - 2)(\alpha - 3)} = \frac{24^3}{24} = 576 \\ \gamma_1 &= \frac{576 - 3(48)(6) + 2(6^3)}{12^{1.5}} \\ &= \frac{144}{12^{1.5}} = \sqrt{12} = \mathbf{3.4641} \quad (\mathbf{E}) \end{aligned}$$

- 2.2. The mean in 2023 is $\theta/3$. By definition, the 65th percentile is the number π_{65} such that $F(\pi_{65}) = 0.65$, so $F(\theta/3) = 0.65$ for the 2024 version of F . In 2024, F is two-parameter Pareto with inflated parameter $\theta' = (1 + r)\theta$ and $\alpha = 4$, so

$$\begin{aligned} 1 - \left(\frac{\theta'}{\theta' + (\theta/3)} \right)^4 &= 0.65 \\ \frac{(1 + r)\theta}{(1 + r)\theta + \theta/3} &= \sqrt[4]{0.35} \\ \frac{1 + r}{4/3 + r} &= \sqrt[4]{0.35} \\ r(1 - \sqrt[4]{0.35}) &= \frac{4}{3}\sqrt[4]{0.35} - 1 \\ r &= \frac{(4/3)\sqrt[4]{0.35} - 1}{1 - \sqrt[4]{0.35}} = \mathbf{0.1107} \end{aligned}$$

- 2.3. To scale a lognormal, you leave σ and add the logarithm of the scale to μ . So already, B and D are the only possibilities, and the question is whether you add or subtract $\ln 1.3$. But clearly the mean in dollars is higher than the mean in euros, so you add $\ln 1.3$ and get **(D)**.
- 2.4. Let X be the original variable, $Z = 1.2X$. Since the mean is 25,000, the parameter θ is $25,000(\alpha - 1) = 50,000$.

$$\begin{aligned} \Pr(X > 100,000) &= \left(\frac{50}{150} \right)^3 = \frac{1}{27} \\ \Pr(Z > 100,000) &= \left(\frac{60}{160} \right)^3 = \frac{27}{512} \\ \frac{27}{512} - \frac{1}{27} &= \mathbf{0.0157} \quad (\mathbf{A}) \end{aligned}$$

- 2.5. The key is to understand (iii). If (for example) 30% of losses exceed \$10000, what percentage does not exceed \$10000? (Answer: 70%) And what percentile of the distribution of losses is \$10000? (Answer: 70th). So statement (iii) is saying that the $100(1 - p)$ th percentile of losses in 1996 equals the mean of losses in 1997. Got it?

The mean of 1997 losses is $\exp(\mu + \ln k + \frac{\sigma^2}{2})$. The $100(1 - p)$ th percentile is $\exp(\mu - z_p\sigma)$. So:

$$\mu - z_p\sigma = \mu + \ln k + \frac{\sigma^2}{2}$$

$$\frac{\sigma^2}{2} + \sigma z_p + \ln k = 0$$

$$\sigma = -z_p \pm \sqrt{z_p^2 - 2 \ln k} \quad (\text{B})$$

Notice that p must be less than 0.5, by the following reasoning. In general, the median of a lognormal (e^μ) is less than (or equal to, if $\sigma = 0$) the mean ($e^{\mu+\sigma^2/2}$), so the median of losses in 1996 is no more than the mean of losses in 1996, which in turn is less than the mean of losses in 1997 since $k > 1$, so $100p$ must be less than 50. Since p is less than 0.5, it follows that z_p will be negative, and σ is therefore positive, as it should be.

- 2.6. We recognize the 1993 distribution as a single-parameter Pareto with $\theta = 1$, $\alpha = 3$. The inflated parameters are $\theta = 1.1$, $\alpha = 3$. $\left(\frac{1.1}{2.2}\right)^3 = \mathbf{0.125}$. (C)
- 2.7. Let X be the inflated variable, with $\theta = 525$, $\alpha = 1.5$. $\Pr(X > 200) = \left(\frac{525}{525+200}\right)^{1.5} = 0.6162$. Let F be the original distribution function, F^* the distribution of $X \mid X > 200$. Then $F(200) = 1 - 0.6162 = 0.3838$ and

$$F^*(x) = \Pr(X \leq x \mid X > 200) = \frac{\Pr(200 < X \leq x)}{\Pr(X > 200)} = \frac{F(x) - F(200)}{1 - F(200)}$$

So to calculate the median, we set $F^*(x) = 0.5$, which means

$$\frac{F(x) - F(200)}{1 - F(200)} = 0.5$$

$$\frac{F(x) - 0.3838}{0.6162} = 0.5$$

$$F(x) = 0.5(0.6162) + 0.3838 = 0.6919$$

We must find x such that $F(x) = 0.6919$.

$$1 - \left(\frac{525}{525+x}\right)^{1.5} = 0.6919$$

$$\frac{525}{525+x} = 0.4562$$

$$\frac{525 - 525(0.4562)}{0.4562} = x$$

$$x = \mathbf{625.87} \quad (\text{C})$$

- 2.8. All the distributions are parameterized so that θ is the scale parameter and is multiplied by $1+i$; no other parameters change, and you should never divide by $1+i$. Therefore **only 1** is correct. (A)
- 2.9. The Pareto is a scale distribution with scale parameter θ , so annual inflation of 10% increases θ by 10%, making it 440,000. The proportion of claims above 750,000 is $S(750,000) = \left(\frac{\theta}{\theta+750,000}\right)^\alpha$. Hence, the proportion this year is $\left(\frac{400,000}{400,000+750,000}\right)^2 = 0.120983$ and the proportion next year is $\left(\frac{440,000}{440,000+750,000}\right)^2 = 0.136714$. The ratio is $\frac{0.136714}{0.120983} = \mathbf{1.1300}$. (D)

- 2.10. This is:

$$\frac{\left(\frac{1.06\theta}{1.06\theta+d}\right)^2}{\left(\frac{\theta}{\theta+d}\right)^2} = \frac{1.06^2(\theta+d)^2}{(1.06\theta+d)^2} \rightarrow 1.06^2 = \mathbf{1.1236}. \quad (\text{C})$$

- 2.11. If X is normal, then $aX + b$ is normal as well. In particular, $1.05X$ is normal. So the distribution of claims after 5% uniform inflation is normal.

For any distribution, multiplying the distribution by a constant multiplies the mean and standard deviation by that same constant. Thus in this case, the new mean is 1050 and the new standard deviation is 105. (E)

- 2.12. Add $\ln 1.1$ to μ : $17.953 + \ln 1.1 = 18.048$. σ does not change. (B)

- 2.13. The k^{th} moment for an exponential is given in the tables:

$$\mathbf{E}[X^k] = k!\theta^k$$

for $k = 4$ and the mean $\theta = 10$, this is $4!(10^4) = \mathbf{240,000}$.

- 2.14. While Y is Weibull, you don't need to know that. It's simpler to use $Y^2 = X^3$ and look up the third moment of an exponential.

$$\mathbf{E}[X^3] = 3!\theta^3 = 6(2^3) = \mathbf{48}$$

- 2.15. We calculate $\mathbf{E}[Y]$ and $\mathbf{E}[Y^2]$, or $\mathbf{E}[X^{-1}]$ and $\mathbf{E}[X^{-2}]$. Note that the special formula in the tables for integral moments of a gamma, $\mathbf{E}[X^k] = \theta^k(\alpha + k - 1) \cdots \alpha$ only applies when k is a *positive* integer, so it cannot be used for the -1 and -2 moments. Instead, we must use the general formula for moments given in the tables,

$$\mathbf{E}[X^k] = \frac{\theta^k \Gamma(\alpha + k)}{\Gamma(\alpha)}$$

For $k = -1$, this is

$$\mathbf{E}[X^{-1}] = \frac{\theta^{-1} \Gamma(\alpha - 1)}{\Gamma(\alpha)} = \frac{1}{\theta(\alpha - 1)}$$

since $\Gamma(\alpha) = (\alpha - 1)\Gamma(\alpha - 1)$. For $k = -2$,

$$\mathbf{E}[X^{-2}] = \frac{1}{\theta^2(\alpha - 1)(\alpha - 2)}$$

Therefore,

$$\text{Var}(Y) = \frac{1}{10^2(1.5)(0.5)} - \left(\frac{1}{10(1.5)} \right)^2 = \mathbf{0.00888889}$$



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Lesson 3

Variance

You learned everything in this lesson in your probability course. Nevertheless, many students miss a lot of these points. They are very important.

There won't necessarily be any exam questions testing you directly on this material (although there could be). Rather, this material is background needed for the rest of the course.

3.1 Additivity

Expected value is **linear**, meaning that $E[aX + bY] = aE[X] + bE[Y]$, regardless of whether X and Y are independent or not. Thus $E[(X + Y)^2] = E[X^2] + 2E[XY] + E[Y^2]$, for example. This means that $E[(X + Y)^2]$ is **not** equal to $E[X^2] + E[Y^2]$, unless $E[XY] = 0$.

Also, it is **not** true in general that $E[g(X)] = g(E[X])$. So $E[X^2] \neq (E[X])^2$.

Since **variance** can be expressed in terms of expected value as $\text{Var}(X) = E[X^2] - E[X]^2$, this allows us to develop a formula for $\text{Var}(aX + bY)$. If you work it out, you get

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + 2ab \text{Cov}(X, Y) + b^2 \text{Var}(Y) \quad (3.1)$$

In particular, if $\text{Cov}(X, Y) = 0$ (which is true if X and Y are independent), then

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

and generalizing to n independent variables,

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i)$$

If all the X_i 's are independent and have identical distributions, and we set $X = X_i$ for all i , then

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = n \text{Var}(X)$$

However, $\text{Var}(nX) = n^2 \text{Var}(X)$, not $n \text{Var}(X)$. You must distinguish between these two situations, which are quite different. Think of the following example. The stock market goes up or down randomly each day. We will assume that each day's change is independent of the previous day's, and has the same distribution. Compare the variance of the following possibilities:

1. You put \$1 in the market, and leave it there for 10 days.
2. You put \$10 in the market, and leave it there for 1 day.

In the first case, there are going to be potential ups and downs each day, and the variance of the change of your investment will be 10 times the variance of one day's change because of this averaging. In the second case, however, you are magnifying a single day's change by 10—there's no dampening of the change by 10 different independent random events, the change depends on a single random event. Therefore the variance is multiplied by 100.

In the more general case where the variables are not independent, you need to know the covariance. This can be provided in a **covariance matrix**. If you have n random variables $X_1, \dots, X_n$, this $n \times n$ matrix $\mathbf{A}$ has $a_{ij} = \text{Cov}(X_i, X_j)$ for $i \neq j$. For $i = j$, $a_{ii} = \text{Var}(X_i)$. This matrix is symmetric and positive semi-definite. However, the covariance of two random variables may be negative.

EXAMPLE 3A For a loss X on an insurance policy, let X_1 be the loss amount and X_2 the loss adjustment expenses, so that $X = X_1 + X_2$. The covariance matrix for these random variables is

$$\begin{pmatrix} 25 & 5 \\ 5 & 2 \end{pmatrix}$$

Calculate the variance in total cost of a loss including loss adjustment expenses. ■

SOLUTION: In formula (3.1), $a = b = 1$, so $25 + 2(5) + 2 = \boxed{37}$. □

A **sample** is a set of observations from n independent identically distributed random variables. The **sample mean** $\bar{X}$ is the sum of the observations divided by n . The variance of the sample mean of $X_1, \dots, X_n$, which are observations from the random variable X , is

$$\text{Var}(\bar{X}) = \text{Var}\left(\frac{\sum_{i=1}^n X_i}{n}\right) = \frac{n \text{Var}(X)}{n^2} = \frac{\text{Var}(X)}{n} \quad (3.2)$$

3.2 Normal approximation

The **Central Limit Theorem** says that for any distribution with finite variance, the sample mean of a set of independent identically distributed random variables approaches a normal distribution. By the previous section, the mean of the sample mean of observations of X is $E[X]$ and the variance is σ^2/n . These parameters uniquely determine the **normal distribution** that the sample mean converges to. A random variable Y with normal distribution with mean μ and variance σ^2 can be expressed in terms of a **standard normal random variable** Z in the following way:

$$Y = \mu + \sigma Z$$

and you can obtain the distribution of Z using the normal distribution calculator you get at the exam.

The normal approximation consists of calculating percentiles of a random variable by assuming that it has a normal distribution. Let $\Phi(x)$ be the cumulative distribution function of the standard normal distribution. (The standard normal distribution has $\mu = 0$, $\sigma = 1$. Φ is the symbol generally used for this distribution function.) Suppose we are given that X is a normal random variable with mean μ , variance σ^2 ; we will write $X \sim n(\mu, \sigma^2)$ to describe X . And suppose we want to calculate the 95th **percentile** of X ; in other words, we want a number x such that $\Pr(X \leq x) = 0.95$. We would reason as follows:

$$\begin{aligned} \Pr(X \leq x) &= 0.95 \\ \Pr\left(\frac{X - \mu}{\sigma} \leq \frac{x - \mu}{\sigma}\right) &= 0.95 \\ \Phi\left(\frac{x - \mu}{\sigma}\right) &= 0.95 \\ \frac{x - \mu}{\sigma} &= \Phi^{-1}(0.95) \\ x &= \mu + \sigma \Phi^{-1}(0.95) \end{aligned}$$

Note that $\Phi^{-1}(0.95) = 1.645$ is a commonly used percentile of the normal distribution.

You should internalize the above reasoning so you don't have to write it out each time. Namely, to calculate a percentile of a random variable being approximated normally, find the value of x such that $\Phi(x)$ is that percentile. Then **scale** x : multiply by the standard deviation, and then **translate** x : add the mean.

This method will be used repeatedly throughout the course.

EXAMPLE 3B A big fire destroyed a building in which 100 of your insureds live. Each insured has a fire insurance policy. The losses on this policy follow a Pareto distribution with $\alpha = 3$, $\theta = 2000$. Even though all the insureds live in the same building, the losses are independent. You are now setting up a reserve for the cost of these losses.

Using the normal approximation, calculate the size of the reserve you should put up if you want to have a 95% probability of having enough money in the reserve to pay all the claims. ■

SOLUTION: The mean of each loss is $\frac{2000}{2} = 1000$ and the variance is

$$E[X^2] - E[X]^2 = \frac{2(2000^2)}{2} - 1000^2 = 3,000,000$$

The mean of the sum is the sum of the means, or $(100)(1000) = 100,000$. The variance of the sum is the sum of the variances, or $100(3,000,000) = 3 \times 10^8$. $\sigma = \sqrt{3 \times 10^8} = 17,320.51$. For a standard normal distribution, the 95th percentile is 1.645. We scale this by 17,320.51 and translate it by 100,000: $100,000 + 17,320.51(1.645) = \boxed{128,492.24}$. $\square$

The normal approximation is sometimes called the *large sample* estimate, since it is based on the Central Limit Theorem, which describes the behavior of the distribution of a sample as its size goes to infinity.

Continuity correction

When a discrete distribution is approximated with the normal distribution, a **continuity correction** is required. If the discrete distribution can assume values a and b but cannot assume values in between a and b , and you want the probability of being strictly above a , you estimate it with the probability that the normal variable is greater than $(a + b)/2$. Use the complement of that if you want the probability of being less than or equal to a . The same goes for b ; if you want the probability of being strictly less than b (notice that this is identical to the probability that the variable is less than or equal to a), use the probability that the normal variable is less than $(a + b)/2$. If you want the probability that the variable is greater than or equal to b (which is identical to the probability that the variable is strictly greater than a), use the probability that the normal random variable is greater than $(a + b)/2$.

EXAMPLE 3C  The distribution of loss sizes is

Size	Probability
1000	0.50
1500	0.25
2500	0.25

Calculate the probability that the average of 100 losses is less than 1550 using the normal approximation. $\blacksquare$

SOLUTION: The mean is 1500. The variance is the second moment minus the mean squared:

$$\sigma^2 = 0.5(1000^2) + 0.25(1500^2 + 2500^2) - 1500^2 = 375,000$$

The variance of the sample mean is the variance of the distribution divided by 100, or 3750.

Losses are always multiples of 500. Adding up 100 of them and dividing by 100, the average is a multiple of 5. Therefore, to calculate the probability of the average being less than 1550, we calculate the probability that the normal variable is less than 1547.5, the midpoint between the possible values for the mean, 1545 and 1550.

$$\Pr(X < 1547.5) = \Phi\left(\frac{1547.5 - 1500}{\sqrt{3750}}\right) = \Phi(0.77567) = \boxed{0.78103}$$

$\square$

3.3 Bernoulli shortcut



A **Bernoulli distribution** is one where the random variable is either 0 or 1. It is 1 with probability q and 0 with probability $1 - q$. Its mean is q , and its variance is $q(1 - q)$. These are in your tables, but you'll use these so often you shouldn't have to look it up.



However, any random variable which can only assume two values is a scaled and translated Bernoulli. If X is Bernoulli and Y can only assume the values a and b , with a having probability q , then $Y = (a - b)X + b$. This means that the **variance of Y** is $(a - b)^2 \text{Var}(X) = (a - b)^2 q(1 - q)$. Remember this!

To repeat, for any random variable which assumes only two values, the variance is the squared difference between the two values, times the probabilities of the two values. This is faster than slogging through a calculation of $E[X]$ and $E[X^2]$.

OK, so quickly—if a random variable is equal to 20 with probability 0.7 and 120 with probability 0.3, what's its variance? (Answer below¹)

This shortcut will be used repeatedly throughout the course.

EXAMPLE 3D For a one-year term life insurance policy of 1000:

- i) The premium is 30.
- ii) The probability of death during the year is 0.02.
- iii) The company has expenses of 2.
- iv) If the insured survives to the end of the year, the company pays a dividend of 3.

Ignore interest.

Calculate the variance in the amount of profit the company makes on this policy. ■

SOLUTION: There are only two possibilities—either the insured dies or he doesn't—so we have a Bernoulli here. We can ignore premium and expenses, since these don't vary, so they generate no variance. Either the company pays 1000 (probability 0.02) or it pays 3 (probability 0.98). The variance is therefore

$$(1000 - 3)^2(0.02)(0.98) = \boxed{19,482.5764}.$$

□

A random variable which can only assume one of two values is called a *two point mixture*. We will learn about mixtures in Section 4.1.

Exercises

3.1. [4B-S93:9] (1 point) If X and Y are independent random variables, which of the following statements are true?

- 1. $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$
- 2. $\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y)$
- 3. $\text{Var}(aX + bY) = a^2 E[X^2] - a(E[X])^2 + b^2 E[Y^2] - b(E[Y])^2$

- (A) 1 (B) 1,2 (C) 1,3 (D) 2,3 (E) 1,2,3

3.2. [4B-F95:28] (2 points) Two numbers are drawn independently from a uniform distribution on $[0,1]$.

What is the variance of their product?

- (A) 1/144 (B) 3/144 (C) 4/144 (D) 7/144 (E) 9/144

¹ Answer: $(1000 - 3)^2(0.02)(0.98) = 19,482.5764$

- 3.3. [4B-F99:7] (2 points) A player in a game may select one of two fair, six-sided dice. Die A has faces marked with 1, 2, 3, 4, 5 and 6. Die B has faces marked with 1, 1, 1, 6, 6, and 6. If the player selects Die A, the payoff is equal to the result of one roll of Die A. If the player selects Die B, the payoff is equal to the mean of the results of n rolls of Die B. The player would like the variance of the payoff to be as small as possible. Determine the smallest value of n for which the player should select Die B.
- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5
- 3.4. [151-82-92:4] A company sells group travel-accident life insurance with b payable in the event of a covered individual's death in a travel accident. The gross premium for a group is set equal to the expected value plus the standard deviation of the group's aggregate claims. The standard premium is based on the following assumptions:
- i) All individual claims within the group are mutually independent; and
 - ii) $b^2q(1 - q) = 2500$, where q is the probability of death by travel accident for an individual.
- In a certain group of 100 lives, the independence assumption fails because three specific individuals always travel together. If one dies in an accident, all three are assumed to die. Determine the difference between this group's premium and the standard premium.
- (A) 0 (B) 15 (C) 30 (D) 45 (E) 60
- 3.5. You are given the following information about the random variables X and Y :
- i) $\text{Var}(X) = 9$
 - ii) $\text{Var}(Y) = 4$
 - iii) $\text{Var}(2X - Y) = 22$
- Determine the correlation coefficient of X and Y .
- (A) 0 (B) 0.25 (C) 0.50 (D) 0.75 (E) 1
- 3.6. [151-82-93:9] (1 point) For a health insurance policy, trended claims will be equal to the product of the claims random variable X and a trend random variable Y . You are given:
- i) $E[X] = 10$
 - ii) $\text{Var}(X) = 100$
 - iii) $E[Y] = 1.20$
 - iv) $\text{Var}(Y) = 0.01$
 - v) X and Y are independent
- Determine the variance of trended claims.
- (A) 144 (B) 145 (C) 146 (D) 147 (E) 148
- 3.7. X and Y are two independent exponentially distributed random variables. You are given that $\text{Var}(X) = 25$ and $\text{Var}(XY) = 7500$. Determine $\text{Var}(Y)$.
- (A) 25 (B) 50 (C) 100 (D) 200 (E) 300

Solutions

- 3.1. The first and second statements are true by formula (3.1). The third statement should have squares on the second a and second b , since

$$\text{Var}(aX) = \mathbf{E}[(aX)^2] - \mathbf{E}[aX]^2 = a^2 \mathbf{E}[X^2] - a^2 \mathbf{E}[X]^2$$

for example. (B)

- 3.2. The mean of the uniform distribution is $\frac{1}{2}$ and the second moment is $\frac{1}{3}$. So

$$\begin{aligned} \text{Var}(XY) &= \mathbf{E}[X^2Y^2] - \mathbf{E}[X]^2 \mathbf{E}[Y]^2 \\ &= \left(\frac{1}{3}\right)\left(\frac{1}{3}\right) - \left(\frac{1}{4}\right)\left(\frac{1}{4}\right) \\ &= \frac{1}{9} - \frac{1}{16} = \frac{7}{144} \quad (\text{D}) \end{aligned}$$

- 3.3. The variance of Die A is

$$\frac{1}{6} \left(\left(1 - \frac{7}{2}\right)^2 + \left(2 - \frac{7}{2}\right)^2 + \left(3 - \frac{7}{2}\right)^2 + \left(4 - \frac{7}{2}\right)^2 + \left(5 - \frac{7}{2}\right)^2 + \left(6 - \frac{7}{2}\right)^2 \right) = \frac{35}{12}$$

Die B is Bernoulli, with only two possible values of 1 and 6 with probabilities $\frac{1}{2}$, so the variance of one toss is $5^2\left(\frac{1}{2}\right)^2 = \frac{25}{4}$. The variance of the mean is the variance of one toss over n (equation (3.2)). So

$$\begin{aligned} \frac{25}{4n} &< \frac{35}{12} \\ 140n &> 300 \\ n &> 300/140 > 2 \end{aligned}$$

The answer is **3**. (C)

- 3.4. The number of fatal accidents for each life, N , has a Bernoulli distribution with mean q and variance $q(1 - q)$, so the variance in one life's aggregate claims is the variance of bN . $\text{Var}(bN) = b^2 \text{Var}(N) = b^2 q(1 - q) = 2500$. For 100 independent lives, aggregate claims are $100bN$, with variance $100 \text{Var}(bN) = 100(2500)$.

For three lives always traveling together, aggregate claims are $3bN$ with variance $3^2 \text{Var}(bN) = 9(2500)$. If we add this to the variance of aggregate claims for the other 97 independent lives, the variance is $9(2500) + 97(2500) = 106(2500)$. The expected value of aggregate claims, however, is no different from the expected value of the totally independent group's aggregate claims.

The difference in premiums is therefore

$$\sqrt{106(2500)} - \sqrt{100(2500)} = \mathbf{14.7815} \quad (\text{B})$$

- 3.5.

$$22 = \text{Var}(2X - Y) = 4(9) + 4 - 2(2) \text{Cov}(X, Y)$$

$$\text{Cov}(X, Y) = 4.5$$

$$\rho_{XY} = \frac{4.5}{\sqrt{9}\sqrt{4}} = \mathbf{0.75} \quad (\text{D})$$

- 3.6.

$$\mathbf{E}[XY] = (10)(1.20) = 12$$

$$\mathbf{E}[(XY)^2] = (\mathbf{E}[X^2]) (\mathbf{E}[Y^2]) = (10^2 + 100)(1.20^2 + 0.01) = 290$$

$$\text{Var}(XY) = 290 - 12^2 = \mathbf{146} \quad (\text{C})$$

- 3.7. For an exponential variable, the variance is the square of the mean. Let θ be the parameter for Y

$$\begin{aligned}\text{Var}(XY) &= \mathbf{E}[X^2] \mathbf{E}[Y^2] - \mathbf{E}[X]^2 \mathbf{E}[Y]^2 \\ 7500 &= (25 + 25)(2\theta^2) - 25\theta^2 \\ &= 75\theta^2 \\ \theta &= 10 \\ \text{Var}(Y) = \theta^2 &= \boxed{100} \quad (\text{C})\end{aligned}$$



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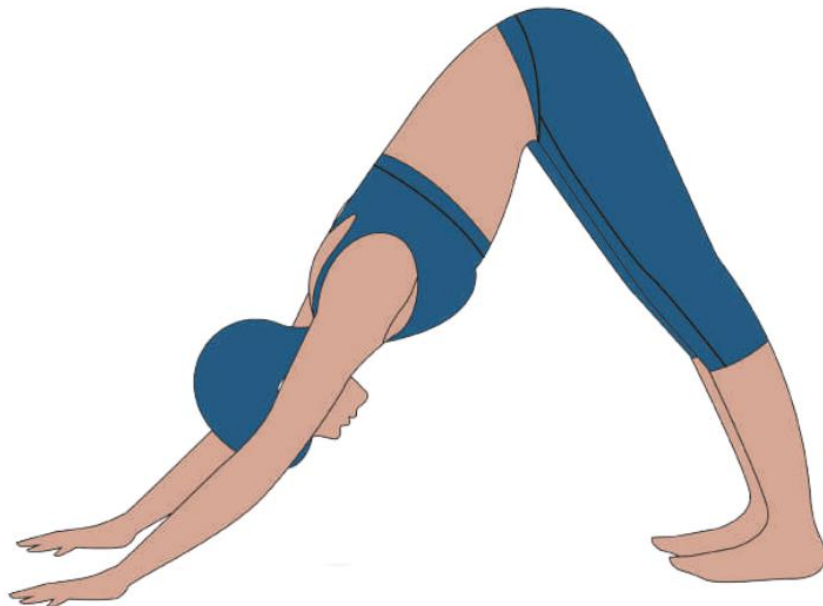
STUDY BREAK



Do you have any fun ideas
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Exercise is a proven way to boost your verbal memory, thinking, and learning. ^[1]



Downward Facing Dog (var 1)

Yoga - for your Back

*Hold each pose for three long breaths, breathing in through
your nose and out through your mouth.

Note that we are mathematicians, not yogi: attempt poses at your own risk!

[1]: <https://www.health.harvard.edu/blog/regular-exercise-changes-brain-improve-memory-thinking-skills-201404097110>

a/s/m

Actuarial Study Materials

Lesson 4

Mixtures and the Conditional Variance Formula

Reading: *Loss Models* Fifth Edition 4.2.3, 5.2.4, 5.2.6

4.1 Mixtures

4.1.1 Discrete mixtures

A (finite) **mixture distribution** is a random variable X whose distribution function can be expressed as a weighted average of n distribution functions of random variables $X_i, i = 1, \dots, n$. In other words,

$$F_X(x) = \sum_{i=1}^n w_i F_{X_i}(x)$$

with the weights $w_i \geq 0$ adding up to 1. Since the density function is the derivative of the distribution function, the density function is the same weighted average of the individual density functions:

$$f_X(x) = \sum_{i=1}^n w_i f_{X_i}(x)$$

If discrete variables are mixed, the probabilities of the mixture are the weighted averages of the component probabilities.

For example, suppose X is a mixture of an **exponential distribution** with mean 100 and weight 60% and an exponential distribution with mean 200 and weight 40%. Then the probability that $X \leq 100$ is

$$\Pr(X \leq 100) = 0.6(1 - e^{-100/100}) + 0.4(1 - e^{-100/200}) = 0.6(0.6321) + 0.4(0.3935) = \boxed{0.5367}$$

A mixture is not the same as a sum of random variables! The distribution function for a sum of random variables—even when they are identically distributed—is usually difficult to calculate. It is important not to confuse the two. Let's consider the situations where each would be appropriate.

A **sum of random variables** is an appropriate model for a situation where several distinct events occur, and you are interested in the sum. Each event may have the same distribution, or may not. Examples of sums of random variables are:

1. The total of a random sample of n items is a sum of random variables. For a random sample, the items are independent and identically distributed.
2. Aggregate loss on a policy with multiple coverages is a sum of random variables, one for each coverage. If a homeowner's policy has coverage for fire, windstorm, and theft, the aggregate loss for each of these three coverages could have its own distribution X_i , and then the aggregate loss for the entire policy would be $X_1 + X_2 + X_3$. These distributions would be different, and may or may not be independent.

A mixture distribution is an appropriate model for a situation where a single event occurs. However, the single event may be of many different types, and the type is random. For example, let X be the cost of a dental claim. This is a single claim and X has a distribution function. However, this claim could be for **preventative work** (cleaning

and scaling), **basic services** (fillings), or **major services** (crowns). Each type of work has a distribution X_i . If 40% of the claims are for preventative work, 35% for basic services, and 25% for major services, then the distribution of X will be a weighted average of the distributions of the 3 X_i 's: $F_X(x) = 0.4F_{X_1}(x_1) + 0.35F_{X_2}(x_2) + 0.25F_{X_3}(x_3)$. It is **not** true that $X = 0.4X_1 + 0.35X_2 + 0.25X_3$. In fact, the type of work that occurred is random; it is not true that every claim is 40% preventative, 35% basic, and 25% major. If that were true, there would be less variance in claim size!

Since a mixture is a single random variable, it can be used as a model even when there is no justification as given in the last paragraph, if it fits the data well.

For calculating means, the **mean of a mixture** is the weighted average of the means of the components. Since the densities are weighted averages and the expected values are integrals of densities, the expected value of a mixture is the weighted average of the expected values of the components. This is true for any raw moment, not just the first moment. But this immediately implies that the **variance of a mixture** is **not** the weighted average of the variances of its components. You must compute the variance by computing the second moment and then subtracting the square of the mean.

EXAMPLE 4A Losses on an auto liability coverage follow a distribution that is a mixture of two Paretos. Each distribution in the mixture has equal weight. One distribution has parameters $\alpha = 3$ and $\theta = 1000$, and the other has parameters $\alpha = 3$ and $\theta = 10,000$.

Calculate the variance of a loss.

SOLUTION: Let X be loss size. We have

$$\begin{aligned} E[X] &= 0.5\left(\frac{1000}{2}\right) + 0.5\left(\frac{10,000}{2}\right) = 2750 \\ E[X^2] &= 0.5(1000^2) + 0.5(10,000^2) = 50,500,000 \\ \text{Var}(X) &= E[X^2] - E[X]^2 = 50,500,000 - 2750^2 = \boxed{42,937,500} \end{aligned}$$

EXAMPLE 4B The severity distribution for losses on an auto collision coverage is as follows:

$$F(x) = 1 - 0.7\left(\frac{2000}{2000 + x}\right)^3 - 0.3\left(\frac{7500}{7500 + x}\right)^4 \quad x \geq 0$$

Calculate the coefficient of variation of loss size.

SOLUTION: This distribution is a mixture of two Pareto distributions, the first with parameters $\alpha = 3$, $\theta = 2000$ and the second with $\alpha = 4$, $\theta = 7500$. We calculate the first two moments:

$$\begin{aligned} E[X] &= 0.7\left(\frac{2000}{2}\right) + 0.3\left(\frac{7500}{3}\right) = 1450 \\ E[X^2] &= 0.7\left(\frac{2(2000)^2}{(2)(1)}\right) + 0.3\left(\frac{2(7500)^2}{(3)(2)}\right) = 8,425,000 \\ \text{Var}(X) &= E[X^2] - E[X]^2 = 8,425,000 - 1450^2 = 6,322,500 \end{aligned}$$

It then follows that the coefficient of variation is

$$CV = \frac{\sqrt{6,322,500}}{1450} = \boxed{1.7341}$$

EXAMPLE 4C On an auto collision coverage, there are two classes of policyholders, A and B. 70% of drivers are in class A and 30% in class B. The means and variances of losses for the drivers are:

Class	Mean	Variance
A	300	30,000
B	800	50,000

A claim is submitted by a randomly selected driver.

Calculate the variance of the size of the claim. ■

SOLUTION: This is a mixture situation—a single claim, with probabilities of being one type or another. Let X be claim size.

$$E[X] = 0.7(300) + 0.3(800) = 450$$

$$E[X^2] = 0.7(30,000 + 300^2) + 0.3(50,000 + 800^2) = 291,000$$

$$\text{Var}(X) = 291,000 - 450^2 = \boxed{88,500}$$

□

4.1.2 Continuous mixtures

The material in this subsection, which was on the pre-2022 syllabus, has been removed from the syllabus. But it may be helpful for understanding later material, so it has been retained. Feel free to skip it.

So far we've discussed discrete mixtures. It is also possible for mixtures to be **continuous**. This means that the distribution function of the mixture is an integral of parametric distribution functions of random variables, and a parameter varies according to a distribution function. The latter distribution is called a *mixing distribution*. One example which we'll discuss in detail in Lesson 57 is a random variable based on a **Poisson distribution** with parameter λ , where λ varies according to a **gamma distribution** with parameters α and θ . Here is another example. ■

EXAMPLE 4D ■ The number of losses on a homeowner's policy is binomially distributed with parameters $m = 5$ and q . The parameter q varies by policyholder uniformly between 0 and 0.4.

Calculate the probability of 2 or more losses for a policyholder. ■

SOLUTION: For a single policyholder,

$$\Pr(X = 0 \mid q) = (1 - q)^5$$

$$\Pr(X = 1 \mid q) = 5q(1 - q)^4$$

To calculate the probability for a randomly selected policyholder, we integrate over q using the uniform density function, which here is $\frac{1}{0.4}$, as the weight.

$$\begin{aligned} \Pr(X = 0) &= \frac{1}{0.4} \int_0^{0.4} (1 - q)^5 dq \\ &= -\frac{1}{0.4} \frac{(1 - q)^6}{6} \Big|_0^{0.4} \\ &= \frac{1}{2.4} (1 - 0.6^6) \\ &= \frac{5}{12} (1 - 0.6^6) \\ \Pr(X = 1) &= \frac{5}{0.4} \int_0^{0.4} q(1 - q)^4 dq \end{aligned}$$

This is easier to integrate by substituting $u = 1 - q$.

$$\begin{aligned}
 &= \frac{5}{0.4} \int_{0.6}^1 (1-u)u^4 du \\
 &= 12.5 \left(\frac{u^5}{5} - \frac{u^6}{6} \right) \Big|_{0.6}^1 = 12.5 \left(\frac{1-0.6^5}{5} \right) - 12.5 \left(\frac{1-0.6^6}{6} \right) \\
 &= 2.5(1-0.6^5) - \frac{25}{12}(1-0.6^6) \\
 \Pr(X=0) + \Pr(X=1) &= 2.5(1-0.6^5) - \frac{20}{12}(1-0.6^6) \\
 &= 2.3056 - 1.5889 = 0.7167
 \end{aligned}$$

Therefore, the probability of 2 or more losses is the complement of 0.7167, which is $1 - 0.7167 = 0.2833$. $\square$

4.2 Conditional variance

In Section 1.3, we discussed the **conditional mean formula**. A useful formula for conditional variance can be developed by calculating the second moment and subtracting the first moment squared using that formula:

$$\begin{aligned}
 \text{Var}(X) &= \mathbf{E}[X^2] - \mathbf{E}[X]^2 \\
 &= \mathbf{E}[\mathbf{E}[X^2 | I]] - \mathbf{E}[\mathbf{E}[X | I]]^2 \\
 &= \mathbf{E}[\mathbf{E}[X^2 | I] - \mathbf{E}[X | I]^2] + (\mathbf{E}[\mathbf{E}[X | I]^2] - \mathbf{E}[\mathbf{E}[X | I]]^2) \\
 &= \mathbf{E}[\text{Var}(X | I)] + \text{Var}(\mathbf{E}[X | I])
 \end{aligned}$$

We've derived the conditional variance formula.

Conditional Variance Formula
$\text{Var}_X(X) = \text{Var}_I(\mathbf{E}_X[X I]) + \mathbf{E}_I[\text{Var}_X(X I)]$

(4.1)

This formula is also known as a **double expectation** formula.

This is a very important equation which will be used repeatedly throughout the course. It is especially useful for calculating variance of mixtures. In a mixture, the condition I will indicate the component of the mixture. Let's apply conditional variance to Example 4C.

EXAMPLE 4C (Repeated for convenience) On an auto collision coverage, there are two classes of policyholders, A and B. 70% of drivers are in class A and 30% in class B. The means and variances of losses for the drivers are:

Class	Mean	Variance
A	300	30,000
B	800	50,000

A claim is submitted by a randomly selected driver.
Calculate the variance of the size of the claim. ■

SOLUTION: Let X be claim size. Let the indicator variable I be the class. It has a probability of 0.7 of being class A and 0.3 of being class B. I is Bernoulli, so we'll apply the previous section's shortcut.

$$\begin{aligned}
 \mathbf{E}[\text{Var}(X | I)] &= 0.7(30,000) + 0.3(50,000) = 36,000 \\
 \text{Var}(\mathbf{E}[X | I]) &= (0.7)(0.3)(800 - 300)^2 = 52,500 \\
 \text{Var}(X) &= 36,000 + 52,500 = 88,500
 \end{aligned}$$

Same answer as before. $\square$

EXAMPLE 4E  Claim sizes range between 0 and 1500. The probability that a claim is no greater than 500 is 0.8. Claim sizes are uniformly distributed on $(0, 500]$ and on $(500, 1500]$.

Calculate the coefficient of variation of claim sizes. ■

SOLUTION: Let X be claim size. We will condition on the claim size being greater than or less than 500. Let I be 0 if claim size is in $(0, 500]$ and 1 if claim size is in $(500, 1500]$.

The mean claim size given that it is no greater than 500 is 250, since the mean of a uniform is the midrange. Similarly, mean claim size given that it is greater than 500 is 1000. By the conditional mean formula

$$E[X] = E[E[X | I]] = 0.8(250) + 0.2(1000) = 400$$

For a uniform distribution, the variance is the range squared divided by 12. So the variance of claim size given that it is no greater than 500 is $500^2/12$ and the variance of claim size given that it is greater than 500 is $1000^2/12$. Therefore

$$E[\text{Var}(X | I)] = 0.8(500^2/12) + 0.2(1000^2/12) = 33,333\frac{1}{3}$$

The variance of the expected values can be calculated with the Bernoulli shortcut.

$$\text{Var}(E[X | I]) = (0.8)(0.2)(1000 - 250)^2 = 90,000$$

We conclude that $\text{Var}(X) = 33,333\frac{1}{3} + 90,000 = 123,333\frac{1}{3}$, and the coefficient of variation is $\sqrt{123,333\frac{1}{3}} / 400 = \boxed{0.8780}$.

You could also calculate the variance as the second moment minus the first moment squared. The second moment given that claims are in $(0, 500]$ is $500^2/3$, and the second moment given that claims are in $(500, 1500]$ is the variance of the uniform plus the mean squared, or $1000^2/12 + 1000^2$. Therefore

$$E[X^2] = 0.8(500^2/3) + 0.2(1000^2/12 + 1000^2) = 283,333\frac{1}{3}$$

The variance is $283,333\frac{1}{3} - 400^2 = 123,333\frac{1}{3}$, the same as calculated above. □

Let's do an example where I is continuous.

EXAMPLE 4F  (Same situation as Example 4D.) The number of losses on a homeowner's policy is binomially distributed with parameters $m = 5$ and Q . Q varies by policyholder uniformly between 0 and 0.4.

Calculate the variance of the number of losses for a randomly selected policyholder. ■

SOLUTION: The indicator variable here is Q . Using the *Loss Models* appendix tables, given Q , the expected number of losses is $5Q$ and the variance is $5Q(1 - Q)$. Then $\text{Var}(5Q) = 25 \text{Var}(Q)$, and $E[5Q(1 - Q)] = 5E[Q] - 5E[Q^2]$. For a random variable U having a uniform distribution on $[0, 1]$, the mean is $\frac{1}{2}$, the second moment is $\frac{1}{3}$, and the variance is $\frac{1}{12}$. Since $Q = 0.4U$,

$$E[Q] = E[0.4U] = 0.4\left(\frac{1}{2}\right) = 0.2$$

$$E[Q^2] = E[(0.4U)^2] = 0.16\left(\frac{1}{3}\right) = \frac{16}{300}$$

$$\text{Var}(Q) = \text{Var}(0.4U) = \frac{0.16}{12} = \frac{4}{300}$$

So, letting N be the number of losses random variable,

$$E[\text{Var}(N | Q)] = 5E[Q] - 5E[Q^2] = 1 - \frac{80}{300} = \frac{11}{15}$$

$$\text{Var}(E[N | Q]) = 25\left(\frac{4}{300}\right) = \frac{1}{3}$$

$$\text{Var}(N) = \frac{11}{15} + \frac{1}{3} = \boxed{\frac{16}{15}}$$

□



Quiz 4-1 The random variable S is defined by

$$S = \sum_{i=1}^{50} X_i$$

The variables X_i follow a Poisson distribution with mean θ , and are independent given θ . The parameter θ is uniformly distributed on $[1, 3]$.

Calculate the variance of S .

Exercises

Mixtures

- 4.1.** The random variable X is a mixture of an exponential distribution with mean 5, with weight $\frac{2}{3}$, and an inverse exponential distribution with $\theta = 5$, with weight $\frac{1}{3}$.
Let $H(x)$ be the cumulative hazard function of X .
Calculate $H(3)$.

- 4.2.** You are given the following information about a portfolio of insurance risks:
- There are three classes of risks: A, B, and C.
 - The number of risks in each class, and the mean and standard deviation of claim frequency for each class, are given in the following chart:

Class	Number of Risks	Claim Frequency	
		Mean	Standard Deviation
A	500	0.10	0.20
B	300	0.12	0.25
C	200	0.15	0.35

Determine the standard deviation of claim frequency for a risk randomly selected from the portfolio.

- (A) Less than 0.240
 (B) At least 0.240, but less than 0.244
 (C) At least 0.244, but less than 0.248
 (D) At least 0.248, but less than 0.252
 (E) At least 0.252

- 4.3.  A group of policyholders has three classes. The probability that policyholder is in each class, add mean and standard deviations for each class, are given in the following table.

Probability of class	Mean loss	Standard deviation of loss
0.5	10	12
0.3	20	30
0.2	30	60

The number of claims submitted by each policyholder is identically distributed for all classes.

1000 claims are submitted from this group. For each claim, the class of policyholder submitting the claim is unknown.

Using the normal approximation, calculate x such that there is a 95% probability that the sum of the claims is less than x .

- 4.4.  For a group of 1000 policyholders in three classes, you are given:

Number of policyholders	Mean loss	Standard deviation of loss
500	10	12
300	20	30
200	30	60

The number of claims submitted by each policyholder is identically distributed for all policyholders.

1000 claims are submitted from this group.

Using the normal approximation, calculate x such that there is a 95% probability that the sum of the claims is less than x .

Each policyholder submits one claim.

Using the normal approximation, calculate x such that there is a 95% probability that the sum of the claims is less than x .

- 4.5.  You are given a portfolio of 100 risks in two classes, A and B, each having 50 risks. The losses of the risks in class A have a mean of 10 and a standard deviation of 5. For the entire portfolio, the mean loss is 20 and the standard deviation is 15.

Calculate the standard deviation of losses for risks in class B.

- (A) Less than 9
- (B) At least 9, but less than 13
- (C) At least 13, but less than 17
- (D) At least 17, but less than 21
- (E) At least 21

- 4.6.  Losses for an insurance coverage follow a distribution which is a mixture of an exponential distribution with mean 10 with 75% weight and an exponential distribution with mean 100 with 25% weight.

Calculate the probability that a loss is greater than 50.

- 4.7.  Losses for an insurance coverage follow a distribution which is a mixture of an exponential distribution with mean 5 and an exponential distribution with mean θ . The mean loss size is 7.5. The variance of loss size is 75.

Determine the coefficient of skewness of the loss distribution.

- 4.8.  [M-S05:34, FAM-S Sample Question #46] The distribution of a loss, X , is a two-point mixture:
- i) With probability 0.8, X has a two-parameter Pareto distribution with $\alpha = 2$ and $\theta = 100$.
 - ii) With probability 0.2, X has a two-parameter Pareto distribution with $\alpha = 4$ and $\theta = 3000$.
- Calculate $\Pr(X \leq 200)$.
- (A) 0.76 (B) 0.79 (C) 0.82 (D) 0.85 (E) 0.88
- 4.9.  [M-F06:39] The random variable N has a mixed distribution:
- i) With probability p , N has a binomial distribution with $q = 0.5$ and $m = 2$.
 - ii) With probability $1 - p$, N has a binomial distribution with $q = 0.5$ and $m = 4$.
- Which of the following is a correct expression for $\text{Prob}(N = 2)$?
- (A) $0.125p^2$
(B) $0.375 + 0.125p$
(C) $0.375 + 0.125p^2$
(D) $0.375 - 0.125p^2$
(E) $0.375 - 0.125p$
- 4.10.  [CAS3-F06:20] An insurance company sells hospitalization reimbursement insurance. You are given:
- i) Benefit payment for a standard hospital stay follows a lognormal distribution with $\mu = 7$ and $\sigma = 2$.
 - ii) Benefit payment for a hospital stay due to an accident is twice as much as a standard benefit.
 - iii) 25% of all hospitalizations are for accidental causes.
- Calculate the probability that a benefit payment will exceed \$15,000.
- (A) Less than 0.12
(B) At least 0.12, but less than 0.14
(C) At least 0.14, but less than 0.16
(D) At least 0.16, but less than 0.18
(E) At least 0.18
- 4.11.  A dental policy pays for basic and major dental services. You are given:
- i) 60% of payments are for basic services and 40% are for major services.
 - ii) Claim sizes for basic services follow a Pareto distribution with parameters $\alpha = 1$ and $\theta = 500$.
 - iii) Claim sizes for major medical services follow a Pareto distribution with parameters $\alpha = 1$ and $\theta = 800$.
- Determine the median claim size on this policy.
- (A) 597 (B) 603 (C) 616 (D) 620 (E) 632

- 4.12.  A class takes an exam. Half the students are good and half the students are bad. For good students, grades are distributed according to the probability density function

$$f(x) = \frac{4}{100} \left(\frac{x}{100} \right)^3 \quad 0 \leq x \leq 100$$

For bad students, grades are distributed according to the probability density function

$$f(x) = \frac{2x}{100^2} \quad 0 \leq x \leq 100$$

Determine the median grade on this exam.

The following three exercises are on continuous mixtures. That topic is not on the syllabus, so they may be skipped.

- 4.13.  [151-82-93:11] (2 points) A population is equally divided into two classes of drivers. The number of accidents per individual driver is Poisson for all drivers.

For a driver selected at random from Class 1, the expected number of accidents is uniformly distributed over (0.2, 1.0).

For a driver selected at random from Class 2, the expected number of accidents is uniformly distributed over (0.4, 2.0).

For a driver selected at random from this population, determine the probability of zero accidents.

- (A) 0.41 (B) 0.42 (C) 0.43 (D) 0.44 (E) 0.45

- 4.14.  For a liability coverage, you are given

- i) Losses for each insured follow an exponential distribution with mean γ .
- ii) γ varies by insured.
- iii) γ follows a single-parameter Pareto distribution with parameters $\alpha = 1$ and $\theta = 1000$.

Calculate the probability that a loss will be less than 500.

- 4.15.  [CAS3-F06:19] In 2006, annual claim frequency follows a negative binomial distribution with parameters β and r . β follows a uniform distribution on the interval (0,2) and $r = 4$.

Calculate the probability that there is at least 1 claim in 2006.

- (A) Less than 0.85
- (B) At least 0.85, but less than 0.88
- (C) At least 0.88, but less than 0.91
- (D) At least 0.91, but less than 0.94
- (E) At least 0.94

Conditional variance

- 4.16.  The size of a loss has mean 3λ and variance λ^2 . λ has the following density function:

$$f(x) = 0.00125 \left(\frac{4000}{x} \right)^6 \quad x \geq 4000.$$

Calculate the variance of the loss.

- (A) Less than 22,000,000
- (B) At least 22,000,000, but less than 27,000,000
- (C) At least 27,000,000, but less than 32,000,000
- (D) At least 32,000,000, but less than 37,000,000
- (E) At least 37,000,000

- 4.17.  [4B-S95:14] (3 points) You are given the following:

- For a given risk, the number of claims for a single exposure period will be 1, with probability $3/4$; or 2, with probability $1/4$.
- If only one claim is incurred, the size of the claim will be 80, with probability $2/3$; or 160, with probability $1/3$.
- If two claims are incurred, the size of each claim, independent of the other, will be 80, with probability $1/2$; or 160 with probability $1/2$.

Determine the variance of the pure premium¹ for this risk.

- (A) Less than 3600
- (B) At least 3600, but less than 4300
- (C) At least 4300, but less than 5000
- (D) At least 5000, but less than 5700
- (E) At least 5700

- 4.18.  [4B-F98:8] (2 points) You are given the following:

- A portfolio consists of 75 liability risks and 25 property risks.
- The risks have identical claim count distributions.
- Loss sizes for liability risks follow a Pareto distribution with parameters $\theta = 300$ and $\alpha = 4$.
- Loss sizes for property risks follow a Pareto distribution with parameters $\theta = 1,000$ and $\alpha = 3$.

Determine the variance of the claim size distribution for this portfolio for a single claim.

- (A) Less than 150,000
- (B) At least 150,000, but less than 225,000
- (C) At least 225,000, but less than 300,000
- (D) At least 300,000, but less than 375,000
- (E) At least 375,000

¹For the definition of pure premium, see page 926.

- 4.19. The number of claims on a policy has a Poisson distribution with mean P . P varies by policyholder. P is uniformly distributed on $[1, 2]$.

Calculate the variance of the number of claims.

- (A) $3/2$ (B) $19/12$ (C) $5/3$ (D) $7/4$ (E) $23/12$

- 4.20. Claim size is exponentially distributed with mean λ . λ varies by insured, and follows a Pareto distribution with parameters $\alpha = 5$ and θ . Variance of claim size is 9.75.

Determine θ .

- (A) 4 (B) 5 (C) 6 (D) 7 (E) 8

- 4.21. [4B-F92:23] (2 points) You are given the following:

- A portfolio of risks consists of 2 classes, A and B .
- For an individual risk in either class, the number of claims follows a Poisson distribution.

Class	Number of Exposures	Distribution of Claim Frequency	
		Mean	Standard Deviation
A	500	0.050	0.227
B	500	0.210	0.561
Total Portfolio	1,000		

Determine the standard deviation of the claim frequency for the total portfolio.

- (A) Less than 0.390
 (B) At least 0.390, but less than 0.410
 (C) At least 0.410, but less than 0.430
 (D) At least 0.430, but less than 0.450
 (E) At least 0.450

- 4.22. [1999 C3 Sample:10] An insurance company is negotiating to settle a liability claim. If a settlement is not reached, the claim will be decided in the courts 3 years from now.

You are given:

- There is a 50% probability that the courts will require the insurance company to make a payment. The amount of the payment, if there is one, has a lognormal distribution with mean 10 and standard deviation 20.
- In either case, if the claim is not settled now, the insurance company will have to pay 5 in legal expenses, which will be paid when the claim is decided, 3 years from now.
- The most that the insurance company is willing to pay to settle the claim is the expected present value of the claim and legal expenses plus 0.02 times the variance of the present value.
- Present values are calculated using $i = 0.04$.

Calculate the insurance company's maximum settlement value for this claim.

- (A) 8.89 (B) 9.93 (C) 12.45 (D) 12.89 (E) 13.53

4.23. [151-83-94:6] (2 points) For number of claims N and aggregate claims S , you are given:

- i) $\Pr(N = i) = \frac{1}{3}, i = 0, 1, 2;$
- ii) $E[S | N = 1] = 3;$
- iii) $E[S | N = 2] = 9;$
- iv) $\text{Var}(S | N = 1) = 9;$ and
- v) $\text{Var}(S | N = 2) = 18.$

Determine $\text{Var}(S)$.

- (A) 19 (B) 21 (C) 23 (D) 25 (E) 27

4.24. [C-S07:3] You are given:

- i) Conditional on $Q = q$, the random variables $X_1, X_2, \dots, X_m$ are independent and follow a Bernoulli distribution with parameter q .
- ii) $S_m = X_1 + X_2 + \dots + X_m.$
- iii) The distribution of Q is beta with $a = 1, b = 99$, and $\theta = 1.$

Determine the variance of the marginal distribution of S_{101} .

- (A) 1.00 (B) 1.99 (C) 9.09 (D) 18.18 (E) 25.25

Solutions

4.1. We first write down the survival function:

$$S(x) = \frac{2}{3}(e^{-x/5}) + \frac{1}{3}(1 - e^{-5/x})$$

and then calculate its logarithm at 3:

$$\begin{aligned} H(3) &= -\ln S(3) \\ &= -\ln\left(\frac{2}{3}(e^{-3/5}) + \frac{1}{3}(1 - e^{-5/3})\right) \\ &= -\ln 0.6362 = \boxed{0.4522} \end{aligned}$$

4.2. This is a mixture distribution. The first moment is

$$E[X] = 0.5(0.10) + 0.3(0.12) + 0.2(0.15) = 0.116$$

The second moment is

$$E[X^2] = 0.5(0.10^2 + 0.20^2) + 0.3(0.12^2 + 0.25^2) + 0.2(0.15^2 + 0.35^2) = 0.07707$$

The variance is $0.07707 - 0.116^2 = 0.063614$. The standard deviation is $\sqrt{0.063614} = \boxed{0.252218}$. (E)

4.3. Since there are 1000 random claims “from the group”, this is a sum of 1000 mixture random variables. The random variable for a single claim is a mixture with mean

$$0.5(10) + 0.3(20) + 0.2(30) = 17$$

and second moment

$$0.5(244) + 0.3(1300) + 0.2(4500) = 1412$$

so the variance is $1412 - 17^2 = 1123$. The mean and variance are multiplied by 1000 for 1000 claims, and the normal approximation requires

$$x = 1000(17) + 1.645\sqrt{1000(1123)} = \boxed{18,743.23}$$

- 4.4. Now we have a sum, not a mixture. The mean of the 1000 claims is the same as before, although technically it's calculated as

$$E[X] = 500(10) + 300(20) + 200(30) = 17,000$$

The variance is the sum of the variances, or

$$\text{Var}(X) = 500(12^2) + 300(30^2) + 200(60^2) = 1,062,000$$

The normal approximation requires

$$x = 17,000 + 1.645\sqrt{1,062,000} = \boxed{18,695.23}$$

The variance is lower than in the previous exercise (where it is 1123 per claim, or 1,123,000 for 1000 claims) because the uncertainty on who is submitting the claims has been removed.

- 4.5. Let μ_B be the mean loss for class B, and let σ_B^2 be the variance of loss for class B. Since the classes are equal in size, the raw moments of loss for the entire portfolio are equally weighted raw moments of the losses for each class. The first moment of losses for the entire portfolio is $20 = \frac{1}{2}(10 + \mu_B)$, so $\mu_B = 30$. The second moment of losses for the entire portfolio is

$$\begin{aligned} 20^2 + 15^2 &= \frac{1}{2}(10^2 + 5^2 + 30^2 + \sigma_B^2) \\ 625 &= \frac{1}{2}(1025 + \sigma_B^2) \end{aligned}$$

from which it follows that $\sigma_B = \boxed{15}$. (C)

- 4.6. The probability that a loss is greater than 50 is the weighted average of the probability that each component of the mixture is greater than 50. Let X be a loss. If X_1 is exponential with mean 10 and X_2 is exponential with mean 100, then

$$\begin{aligned} \Pr(X_1 > 50) &= e^{-50/10} = 0.006738 \\ \Pr(X_2 > 50) &= e^{-50/100} = 0.606531 \\ \Pr(X > 50) &= 0.75(0.006738) + 0.25(0.606531) = \boxed{0.1567} \end{aligned}$$

- 4.7. Let w be the weight of the exponential with mean θ . The second moment is $75 + 7.5^2 = 131.25$. From equating the first moments,

$$\begin{aligned} 5(1 - w) + \theta w &= 7.5 \\ 5 - 5w + \theta w &= 7.5 \\ w(\theta - 5) &= 2.5 \end{aligned} \tag{*}$$

From equating the second moments,

$$\begin{aligned} 50(1 - w) + 2w\theta^2 &= 131.25 \\ -50w + 2w\theta^2 &= 81.25 \\ 2w(\theta^2 - 25) &= 81.25 \end{aligned}$$

Dividing the first moment equation into the second moment equation eliminates w :

$$\begin{aligned} 2(\theta + 5) &= \frac{81.25}{2.5} \\ \theta &= \frac{81.25}{2.5(2)} - 5 = 11.25 \end{aligned}$$

Plugging into (*),

$$\begin{aligned} w(6.25) &= 2.5 \\ w &= \frac{2.5}{6.25} = 0.4 \end{aligned}$$

To calculate skewness, we only need $E[X^3]$, since we already know the first and second moments and the variance. For an exponential, $E[X^3] = 6\theta^3$, so

$$\begin{aligned} E[X^3] &= 6(0.6(5^3) + 0.4(11.25^3)) = 3867.1875 \\ \gamma_1 &= \frac{E[X^3] - 3E[X^2]\mu + 2\mu^3}{\sigma^3} \\ &= \frac{3867.1875 - 3(131.25)(7.5) + 2(7.5^3)}{75^{3/2}} \\ &= \frac{3867.1875 - 2953.125 + 843.75}{75^{3/2}} = \boxed{2.70633} \end{aligned}$$

- 4.8. The probability of the mixture is the mixed probability. For a two-parameter Pareto with $\alpha = 2$, $\theta = 100$, the probability is $\Pr(X \leq 200 \mid \text{(i)}) = 1 - \left(\frac{100}{300}\right)^2 = \frac{8}{9}$. For a two-parameter Pareto with $\alpha = 4$, $\theta = 3000$, it is $\Pr(X \leq 200 \mid \text{(ii)}) = 1 - \left(\frac{3000}{3200}\right)^4 = 0.227524$. The mixed probability is $\Pr(X \leq 200) = 0.8(8/9) + 0.2(0.227524) = \boxed{0.756616}$. (A)
- 4.9. $\Pr(N = 2)$ for the first component is $0.5^2 = 0.25$, and for the second component $\binom{4}{2}0.5^4 = 0.375$, so mixing them, $0.25p + 0.375(1 - p) = \boxed{0.375 - 0.125p}$. (E)
- 4.10. This is a mixture distribution. If we let $F_1(x)$ be the standard hospital stay distribution and $F_2(x)$ the accident stay distribution, we add $\ln 2$ to scale the distribution up to $F_2(x)$, so $F_2(x)$ is lognormal with $\mu = 7 + \ln 2$ and $\sigma = 2$. Equivalently, we could calculate $S_2(7500)$. By the Law of Total Probability,

$$\begin{aligned} \Pr(X > 15,000) &= 0.75S_1(15,000) + 0.25S_2(15,000) \\ &= 0.75 \left[1 - \Phi\left(\frac{\ln 15,000 - 7}{2}\right) \right] + 0.25 \left[1 - \Phi\left(\frac{\ln 7,500 - 7}{2}\right) \right] \\ &= 0.75[1 - \Phi(1.31)] + 0.25[1 - \Phi(0.96)] \\ &= 0.75(1 - 0.9049) + 0.25(1 - 0.8315) = \boxed{0.1135} \quad (\text{A}) \end{aligned}$$

- 4.11. We have a mixture of 2 Pareto distributions, and would like to find x such that $F(x) = 0.5$, or equivalently $S(x) = 0.5$.

$$S(x) = 0.6\left(\frac{500}{500 + x}\right) + 0.4\left(\frac{800}{800 + x}\right) = 0.5$$

Multiplying out denominators,

$$\begin{aligned} 300(800) + 300x + 320(500) + 320x &= 0.5(400,000 + 1300x + x^2) \\ 400,000 + 620x &= 0.5x^2 + 650x + 200,000 \\ 0.5x^2 + 30x - 200,000 &= 0 \\ x = -30 + \sqrt{900 + 400,000} &= \boxed{603.167} \quad (\text{B}) \end{aligned}$$

- 4.12. For the overall mixture distribution

$$f(x) = 0.5 \left[\frac{4}{100} \left(\frac{x}{100} \right)^3 \right] + 0.5 \left(\frac{2x}{100^2} \right)$$

$$F(x) = 0.5 \int_0^x f(u) du$$

$$= 0.5 \left[\left(\frac{x}{100} \right)^4 + \left(\frac{x}{100} \right)^2 \right]$$

Setting this equal to 0.5 obtains the median.

$$0.5 \left[\left(\frac{x}{100} \right)^4 + \left(\frac{x}{100} \right)^2 \right] = 0.5$$

Let $y = \left(\frac{x}{100} \right)^2$.

$$y^2 + y - 1 = 0$$

$$y = \frac{-1 + \sqrt{5}}{2} = 0.618034$$

$$x = 100\sqrt{y} = \boxed{78.6151} \quad (\text{D})$$

- 4.13. The probability of 0 accidents is $e^{-\lambda}$. We integrate this over the uniform distribution for each class:

$$\text{Class 1: } \frac{1}{0.8} \int_{0.2}^1 e^{-\lambda} d\lambda = \frac{e^{-0.2} - e^{-1}}{0.8} = 0.5636$$

$$\text{Class 2: } \frac{1}{1.6} \int_{0.4}^2 e^{-\lambda} d\lambda = \frac{e^{-0.4} - e^{-2}}{1.6} = 0.3344$$

The average is $\frac{1}{2}(0.5636 + 0.3344) = \boxed{0.4490}$. (E)

- 4.14. The conditional probability of a loss less than 500 given γ is $F(500 | \gamma) = 1 - e^{-500/\gamma}$. We integrate this over γ , using the density function of a single-parameter Pareto.

$$\Pr(X < 500) = \int_{1000}^{\infty} \left(1 - e^{-500/\gamma} \right) \frac{1000}{\gamma^2} d\gamma$$

We evaluate the integral.

$$\begin{aligned} \Pr(X < 500) &= -\frac{1000}{\gamma} - 2e^{-500/\gamma} \Big|_{1000}^{\infty} \\ &= -2 + \frac{1000}{1000} + 2e^{-500/1000} \\ &= -1 + 2e^{-1/2} = \boxed{0.2131} \end{aligned}$$

- 4.15. If N is the number of claims, $1 - \Pr(N = 0) = 1 - \frac{1}{(1+\beta)^4}$. We integrate this over the mixing uniform distribution, which has density $\frac{1}{2}$.

$$1 - \Pr(N = 0) = \frac{1}{2} \int_0^2 \left(1 - \frac{1}{(1+\beta)^4} \right) d\beta$$

$$\begin{aligned}
&= \frac{1}{2} \left(2 + \left[\frac{1}{3(1+\beta)^3} \right]_0^2 \right) \\
&= 1 + \frac{1}{2} \left(\frac{1}{81} - \frac{1}{3} \right) = \boxed{0.839506} \quad (\text{A})
\end{aligned}$$

- 4.16. We recognize the distribution as a single-parameter Pareto with $\theta = 4000$ and $\alpha = 5$. The mean, $E[\lambda] = \frac{(5)(4000)}{4} = 5000$ and the second moment $E[\lambda^2] = \frac{(5)(4000^2)}{3} = \frac{80,000,000}{3}$. Therefore, the variance of the distribution, $\text{Var}(\lambda) = \frac{80,000,000}{3} - 5000^2 = \frac{5,000,000}{3}$. We will need this information for the formula:

$$\begin{aligned}
\text{Var}(X) &= E[\text{Var}(X | \lambda)] + \text{Var}(E[X | \lambda]) \\
&= E[\lambda^2] + \text{Var}(3\lambda) \\
&= \frac{80,000,000}{3} + (9) \frac{5,000,000}{3} = \boxed{41,666,667} \quad (\text{E})
\end{aligned}$$

- 4.17. Let PP be the pure premium.

$$\text{Var}(PP) = \text{Var}(E[PP | N]) + E[\text{Var}(PP | N)]$$

We will use the Bernoulli shortcut discussed in Section 3.3 on page 52.

$E[PP | N]$ is a random variable. If $N = 1$, it assumes the value of $\frac{2}{3}(80) + \frac{1}{3}(160) = \frac{320}{3}$. When $N = 2$, each claim has expected value $\frac{1}{2}(80) + \frac{1}{2}(160) = 120$, so PP , which is the sum of 2 claims, has expected value $2(120) = 240$. The variance of this random variable is the product of the probabilities of $N = 1$ and $N = 2$ times the square of the difference between $\frac{320}{3}$ and 240:

$$\text{Var}(E[PP | N]) = \left(\frac{3}{4} \right) \left(\frac{1}{4} \right) \left(\frac{320}{3} - 240 \right)^2 = 3333\frac{1}{3}$$

$\text{Var}(PP | N)$ is a random variable. When $N = 1$, the conditional variance is the product of the probabilities of 80 and 160 times the square of the difference between 160 and 80:

$$\text{Var}(PP | N = 1) = \left(\frac{2}{3} \right) \left(\frac{1}{3} \right) (160 - 80)^2 = \frac{12,800}{9}$$

When $N = 2$, the conditional variance of each loss is the product of the probabilities of 80 and 160 times the square of the difference between 160 and 80. Since there are two losses, this is multiplied by 2.

$$\text{Var}(PP | N = 2) = 2 \left(\frac{1}{2} \right) \left(\frac{1}{2} \right) (160 - 80)^2 = 3200$$

The expected value of the $\text{Var}(PP | N)$ is then

$$E[\text{Var}(PP | N)] = \frac{3}{4} \left(\frac{12,800}{9} \right) + \frac{1}{4} (3200) = 1866\frac{2}{3}$$

Putting it all together

$$\text{Var}(PP) = 3333\frac{1}{3} + 1866\frac{2}{3} = \boxed{5200} \quad (\text{D})$$

- 4.18. Let I be the indicator of whether the risk is liability or property.

$$\begin{aligned}\text{Var}(X) &= \text{Var}(\mathbf{E}[X | I]) + \mathbf{E}[\text{Var}(X | I)] \\ \mathbf{E}[X | \text{liability}] &= \frac{300}{4-1} = 100 \\ \text{Var}(X | \text{liability}) &= \frac{2 \cdot 300^2}{3 \cdot 2} - 100^2 = 20,000 \\ \mathbf{E}[X | \text{property}] &= \frac{1,000}{3-1} = 500 \\ \text{Var}(X | \text{property}) &= \frac{2 \cdot 1,000^2}{2 \cdot 1} - 500^2 = 750,000\end{aligned}$$

By the Bernoulli shortcut (Section 3.3, page 52), since the two expected values are 400 apart and have probabilities $\frac{3}{4}$ and $\frac{1}{4}$,

$$\text{Var}(\mathbf{E}[X | I]) = \left(\frac{3}{4}\right)\left(\frac{1}{4}\right)(500 - 100)^2 = 30,000$$

Also,

$$\mathbf{E}[\text{Var}(X | I)] = \left(\frac{3}{4}\right)(20,000) + \left(\frac{1}{4}\right)(750,000) = 202,500$$

So the answer is

$$\text{Var}(X) = 30,000 + 202,500 = \boxed{232,500} \quad (\text{C})$$

- 4.19. $\text{Var}(N) = \text{Var}(\mathbf{E}[N | P]) + \mathbf{E}[\text{Var}(N | P)] = \text{Var}(P) + \mathbf{E}[P] = \frac{1}{12} + \frac{3}{2} = \boxed{\frac{19}{12}}. \quad (\text{B})$

4.20.

$$\begin{aligned}9.75 &= \mathbf{E}[\text{Var}(X | \lambda)] + \text{Var}(\mathbf{E}[X | \lambda]) \\ &= \mathbf{E}[\lambda^2] + \text{Var}(\lambda) \\ &= \frac{2\theta^2}{12} + \left(\frac{2\theta^2}{12} - \frac{\theta^2}{16}\right) = \frac{13}{48}\theta^2 \\ \theta^2 &= 36 \\ \theta &= \boxed{6} \quad (\text{C})\end{aligned}$$

- 4.21. Let I = the random variable indicating the class.

$$\text{Var}(N) = \text{Var}(\mathbf{E}[N | I]) + \mathbf{E}[\text{Var}(N | I)]$$

$\mathbf{E}[N | I]$ is a random variable which assumes the value 0.050 half the time and 0.210 half the time. The probabilities of 0.050 and 0.210 are each 0.50, and the difference of the two values is $0.210 - 0.050 = 0.160$, so by the Bernoulli shortcut, the variance of $\mathbf{E}[N | I]$ is $(0.50)(0.50)(0.160^2) = 0.0064$.

Similarly, $\text{Var}(N | I)$ is a random variable which assumes two values, 0.227^2 and 0.561^2 , each one half the time. The expected value of this random variable is $\frac{1}{2}(0.227^2 + 0.561^2) = 0.183125$.

Putting it all together

$$\begin{aligned}\text{Var}(N) &= 0.0064 + 0.183125 = 0.189525 \\ \sigma_N &= \sqrt{0.189525} = \boxed{0.4353} \quad (\text{D})\end{aligned}$$

This exercise can also be worked out by calculating first and second moments.

$$\begin{aligned}\mathbf{E}[N] &= 0.5(0.05) + 0.5(0.21) = 0.13 \\ \mathbf{E}[N^2] &= 0.5(0.05^2 + 0.227^2) + 0.5(0.21^2 + 0.561^2) = 0.206425 \\ \text{Var}(N) &= 0.206425 - 0.13^2 = 0.189525 \\ \sigma_N &= \sqrt{0.189525} = \boxed{0.4353} \quad (\text{D})\end{aligned}$$

- 4.22. The expected present value of the claim is $0.5(10/1.04^3)$, and the present value of legal fees is $5/1.04^3$, for a total of $10/1.04^3 = 8.89$. We will compute the variance using the conditional variance formula. The legal expenses are not random and have no variance, so we'll ignore them. Let I be the indicator variable for whether a payment is required, and X the settlement value.

$$\text{Var}(X) = \text{Var}(\mathbf{E}[X | I]) + \mathbf{E}[\text{Var}(X | I)]$$

The expected value of the claim is 0 with probability 50% and $10/1.04^3$ with probability 50%. Thus the expected value can only have one of two values. It is a Bernoulli random variable. The Bernoulli shortcut says that its variance is

$$\text{Var}(\mathbf{E}[X | I]) = (0.5)(0.5)\left(\frac{10}{1.04^3}\right)^2 = 19.7579$$

The variance of the claim is 0 with probability 50% and $(20/1.04^3)^2$ with probability 50%. The expected value of the variance is therefore

$$\mathbf{E}[\text{Var}(X | I)] = (0.5)\left(0 + \left(\frac{20}{1.04^3}\right)^2\right) = 158.0629$$

Therefore, $\text{Var}(X) = 19.7579 + 158.0629 = 177.8208$. The answer is

$$8.89 + 0.02(177.8208) = \boxed{12.4464} \quad (\text{C})$$

- 4.23. Naturally, if $N = 0$, both mean and variance are zero.

Using the conditional variance formula,

$$\mathbf{E}[\text{Var}(S | N)] = \frac{0 + 9 + 18}{3} = 9$$

$$\mathbf{E}[\mathbf{E}[S | N]] = \frac{0 + 3 + 9}{3} = 4$$

$$\mathbf{E}[\mathbf{E}[S | N]^2] = \frac{0 + 9 + 81}{3} = 30$$

$$\text{Var}(\mathbf{E}[S | N]) = 30 - 4^2 = 14$$

$$\text{Var}(S) = 9 + 14 = \boxed{23} \quad (\text{C})$$

- 4.24. A common error on this question was to calculate the variance of each X_i and multiply by 101. Since the X_i 's are not unconditionally independent, the variance of S_{101} is not the sum of the variances.

We use the conditional variance formula.

$$\text{Var}(S_{101}) = \mathbf{E}[\text{Var}(S_{101} | Q)] + \text{Var}(\mathbf{E}[S_{101} | Q])$$

$$\mathbf{E}[S_{101} | Q] = 101 \mathbf{E}[X_i | Q] = 101Q$$

$$\text{Var}(S_{101} | Q) = 101 \text{Var}(X_i | Q) = 101Q(1 - Q)$$

$$\mathbf{E}[Q] = \frac{1}{1 + 99} = 0.01$$

$$\mathbf{E}[Q^2] = \frac{a(a+1)}{(a+b)(a+b+1)} = \frac{2}{100(101)}$$

$$\text{Var}(Q) = \frac{ab}{(a+b)^2(a+b+1)} = \frac{99}{100^2(101)}$$

$$\mathbf{E}[101Q(1 - Q)] = 101(\mathbf{E}[Q] - \mathbf{E}[Q^2]) = 101\left(0.01 - \frac{2}{100(101)}\right) = 0.99$$

$$\text{Var}(101Q) = 101^2 \text{Var}(Q) = 0.9999$$

$$\text{Var}(S_{101}) = 0.99 + 0.9999 = \boxed{1.9899} \quad (\text{B})$$

Quiz Solutions

- 4-1. The X_i 's are independent given θ but are not independent, so we can't just add up the individual variances. Instead, we'll use conditional variance, with θ being the condition.

$$\begin{aligned}\mathbf{E}[S \mid \theta] &= 50 \mathbf{E}[X_i \mid \theta] = 50\theta \\ \text{Var}(S \mid \theta) &= 50 \text{Var}(X_i \mid \theta) = 50\theta\end{aligned}$$

Using conditional variance,

$$\begin{aligned}\text{Var}(S) &= \mathbf{E}[\text{Var}(S \mid \theta)] + \text{Var}(\mathbf{E}[S \mid \theta]) \\ &= E_{\theta}(50\theta) + \text{Var}_{\theta}(50\theta) \\ &= 50 \mathbf{E}[\theta] + 2500 \text{Var}(\theta) \\ &= 50(2) + 2500(2^2/12) = \boxed{933\frac{1}{3}}\end{aligned}$$



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Part II

Background Material for Long Term Coverages

Lesson 5

Introduction to Long Term Insurance

Reading: *Actuarial Mathematics for Life Contingent Risks* 3rd edition 1

A long-term insurance contract is a contract in which the insurer makes commitments that extend for more than one year. Generally the insurer offers insurance for a long period of time at a guaranteed premium, or at least limits the degree to which the premium can be changed.

Life insurance is the long-term insurance contract we will concentrate on for most of this course. It is the simplest long-term insurance since the only contingency to consider is death. Any long-term contract has to take death into account; at the very least, death terminates the contract. If other contingencies are involved (like disability or illness), the contract will be more complicated than a life insurance contract.

The first chapter of the textbook is an introduction to long-term insurance. It discusses the features of these insurances, but not the mathematics. The following is a summary of the chapter. While I've tried to include anything important, you may want to read the chapter in the textbook to assure yourself that you have learned every little piece of trivia that could be asked on the exam.^{1 2}

Life insurance pays a lump sum, known as a *sum insured*, upon death. (In the U.S., this amount may also be called the face amount of the policy.) The insured pays a premium at the beginning of the period, often the year, for this insurance. Originally, a group of individuals of a specific age or in a specific age group paid an amount so that the total amount collected equalled the total benefits paid. This is called *assessmentism*. Group life insurance still uses assessmentism. But for individuals, this method means that premiums increase with age. People are likely to withdraw from the group when the premium gets too high.

This led to the level premium insurance contract. In this contract, the same premium is paid each year. These contracts are more complex since the higher premiums collected in the earlier years must be invested to pay for the higher benefits of the later years.

In order to purchase life insurance, one must have *insurable interest* in the life insured. One has to be related to the insured life, or have a close business relationship with that person. Otherwise the insurance is a wager on someone's death, which is not desirable.

Traditional insurance contracts Traditional insurance contracts have fixed premiums and fixed benefits. Life insurance provides a benefit upon death; annuities provide a periodic payment while the annuitant is alive. For life insurance products, premiums are payable periodically (annually, semi-annually, quarterly, or monthly) throughout the life of the contract. However, the premium payment period often is limited to a certain number of years or until the insured reaches a certain age, because older insureds may be less able to pay the premium. For annuities, a single premium may be paid. Otherwise, periodic premiums are payable during a deferral period; the annuity benefits begin after the deferral period. You would not typically have money going both ways at the same time—premiums payable while the annuitant is receiving benefits.

Examples of traditional insurance contracts are

Term insurance *Term insurance* pays a benefit if the policyholder dies within a fixed amount of time. It is useful for insuring a family at a low cost, or insuring a business against losses due to death of an employee. In the latter case, the company pays the premiums and receives the benefits. This type of insurance may be called *key person insurance* or *COLI*, which stands for company owned life insurance.

¹Examples of trivia I've not included are:

1. In the UK, when is it called insurance and when is it called assurance?
2. Which insurance rider inspired a Hollywood feature?

²In this lesson, footnotes are my own comments on the textbook, and occasionally modify the textbook. They are purely to give you the situation in the U.S.; the textbook tends to reflect the international situation. On exams do not use anything from these footnotes!

Other types of term insurance are

1. Decreasing term insurance, which may be used to pay off the balance of a mortgage.
2. **Renewable term insurance** allows renewal at the expiry date, at an increased premium.
3. **Convertible term insurance** allows conversion of the policy to whole life.

Whole life **Whole life insurance** pays a benefit whenever the policyholder dies. Premiums are level. The policy has a cash value that is paid if the policyholder surrenders the policy.

In the U.S., non-forfeiture laws require cash surrender values to be offered on whole life insurance. These cash values increase over the years of the policy, but are less than the face amount.³

Some uses of whole life insurance are:

1. Payment of funeral expenses.
2. Payment of estate taxes.
3. Saving vehicle for younger lives.

Endowment insurance **Endowment insurance** pays a benefit if the policyholder dies within a certain period. If the policyholder survives to the end of the period, the benefit is paid at that point. It has cash values. It is not offered in North America or the U.K. because the investment component does not compete with other investments⁴, but they are popular in developing nations.

Joint life/multiple life insurance This type of insurance pays upon the first death or the last death in a group of individuals. **Joint life** is a policy on husband and wife; companies may insure their officers using multiple life insurance. According to the textbook first-to-die policies are more common than last-to-die policies.⁵

Participating insurance⁶ Whole life insurance must make interest assumptions into the distant future. In order not to lose money, these assumptions are conservatively low. But the product is then unattractive. To compensate, **participating insurance** (par insurance for short) pays non-guaranteed dividends based on the excess of the rate earned on investments over the assumption. These dividends are known as bonuses outside North America.

In North America, policyholders may choose to receive these dividends in many forms, such as

Cash refunds Dividend is paid as cash.

Premium reductions Dividend is offset against premium. This is really no different from a cash refund.

Increased paid up insurance The death benefit is increased.⁷

In the U.K., dividends are used only to increase the death benefit, the third option above. The dividends are called **reversionary bonuses**, and come in three variations⁸:

simple reversionary bonuses Applied to original face amount only

compound reversionary bonuses Applied to sum of original face amount and prior bonuses

super-compound reversionary bonuses Two different bonus rates, one for original face amount, another for prior bonuses.

³If they were greater than the face amount, a policyholder on his death bed would surrender the policy.

⁴And importantly, the tax laws offer very unfavorable treatment to these policies (but the book doesn't mention this).

⁵But in the U.S., last-to-die products on husband/wife are especially popular as a method of paying estate taxes, due to their low premiums. Generally estates pass tax-free to a spouse, so the benefit is not needed until the second death.

⁶For a couple of years, profit tests for participating insurance were on the syllabus. I suspect that material was removed because the book's terminology and description were non-North American, and in fact sound weird to a North American actuary, and the SOA is based in North America. If so, they probably wouldn't test you on this material either.

⁷Other options not mentioned in the textbook are (1) Leave the dividend with the company to accumulate with interest, (2) purchase one-year term insurance; in other words, increase the death benefit for one year only

⁸In the U.S., dividends used to purchase paid up additions follow the "super-compound reversionary bonus" style.

Terminal bonuses⁹ are paid on policies that terminate due to death or surrender.

Advantages/disadvantages of cash dividends versus reversionary bonuses are

1. Cash dividends allow policyholders to use them to pay premiums, which may be helpful if policyholder is short of cash.
2. Cash dividends may be taxable.
3. Cash dividends are simpler to understand.
4. Reversionary bonuses are more consistent with the purpose of the policy.
5. Companies may only give some portion of reversionary bonuses to policyholders who surrender, which is unfair to them.¹⁰
6. Reversionary bonuses make it easier for companies to smooth their dividends.
7. Cash dividends are expensive to operate.
8. Cash dividends require liquidity for the insurer, restricting its investments somewhat.

Additional benefits that policies may provide

Policy loans For policies with cash values, a **policyholder may take a loan against the cash value**. This loan would be deducted against the benefit if it is not repaid before death.

Accelerated death benefits Under this rider, the **death benefit is paid upon terminal illness**.

Accidental death benefit Under this rider, also known as **double indemnity**, **twice the death benefit is paid if death is due to accident**.

Disability waiver Under this rider, the **premium is waived upon disability**.

Family income This rider is term insurance. Under this rider, upon death, **a fixed payment is made each month until the expiry of the rider**. For example, if the rider is for 10 years and the insured dies within 8 years, the policy pays a benefit for 2 years. Thus this is decreasing term insurance. According to the textbook, the base policy is a term insurance for the same number of years.

Reasons the design of life insurance contracts has changed recently

1. Competition with mutual funds and banks for policyholders' savings. Thus insurance products have investment components.
2. Changing demographics and lifecycles.
3. Developments in financial science and technology. Modern financial science improves risk management techniques. Computer technology allow more complex products.
4. Better informed customers. Smarter customers require more competitive products.

Nontraditional insurance contracts Here are examples of non-traditional products. We do not discuss non-traditional products any further in this course

1. **Universal Life**. This product has flexible premiums. The benefit payable upon death may also be adjustable.

⁹In North America they're called terminal dividends.

¹⁰In the U.S., this is not allowed. Once a policyholder gets a paid-up addition, it cannot be taken back. However, companies pay dividends at the end of the policy year (with a provision to pay a partial dividend upon mid-year death); if the policyholder surrenders in the middle of the policy year, they do not get the dividend for that year.

Premiums go into a notional account. It is notional only; there are no designated assets. The insurer subtracts mortality and expense charges and credits interest. The credited interest rate may change from time to time, but cannot be less than 0%. Generally charges and credits are made monthly. The policy stays in force as long as the account balance is positive. In the earlier years, there is a surrender charge if a policyholder surrenders the policy.

2. **Unitized-with-profit.** This is a UK product. In the UK, products that allow policyholders to share in the company's profits (which are called "participating" in the US) are called "with-profit". In unitized-with-profit contracts, the premium purchases units of investments, which earn reversionary bonuses, but cannot decrease in value.
3. Equity-linked insurance. Two types¹¹:
 - (a) **Variable annuity** (called "unit-linked" in the UK) invests premiums in mutual funds. Credited amounts may be positive or negative, depending on investment performance.
 - (b) **Equity-indexed annuity** has guaranteed returns, and the policyholder receives a portion of the growth on an index (e.g. a stock index) if greater than the guarantee.

Distribution methods Most insurance is sold through brokers. Brokers are paid commissions, which are a percentage of premium. Usually the first year commission is a higher percentage of premium than renewal commissions. This is referred to as a *front end load*.

Direct marketing is an alternative to brokers. Typically policies sold by direct marketing are for small face amounts and are not as heavily underwritten as broker-sold policies. Hence mortality is higher, but expenses are lower both due to simplified underwriting and lack of commissions. Often such policies are for funeral expenses or credit insurance—insurance paying off a loan if the insured dies.

Underwriting classes When a company sells insurance, it usually **underwrites** the policyholder; it determines how healthy the policyholder is and sets premiums based on the policyholder's underwriting class. Typically the underwriting classes are:

1. Preferred: low risk factors, no smoking or alcohol abuse, no high-risk occupations or hobbies. Preferred classes are common in North America, but not in the rest of the world.
2. Normal¹²: some risk factors, but insurable at standard rates.
3. Rated: Elevated risk factors, higher than standard rates must be charged.
4. Uninsurable: Too risky, not insurable at any price.

For life insurance, there is adverse selection by policyholders; the sicker one is, the more likely one is to buy life insurance, and for higher amounts. Underwriting therefore is stricter the higher the sum insured is. Underwriting may only require a policyholder statement for a small amount, but will require a physician's statement for a larger amount and may require examination by the company's doctor for very large amounts.

Types of annuities An annuity makes a periodic payment to the purchaser of the annuity. Types of annuities are:

1. Single Premium Deferred Annuity (SPDA). Policyholder pays a single premium. Annuity commences payments on some future date.
2. Single Premium Immediate Annuity (SPIA). Policyholder pays a single premium and annuity payments commence immediately. For example, if payments are monthly, the first payment is one month after the premium is paid.

¹¹For unknown reasons, the textbook discusses annuities here rather than insurances.

¹²Called "standard" in the U.S.

3. Regular Premium Deferred Annuity (RPDA). Policyholder pays premium on a regular basis until annuity payments commence.

These three types categorize annuities based on when the premiums and benefits are paid. The next three types categorize when benefits are paid for annuities on two lives.

4. Joint life annuity. Annuity makes payments only when both are alive.
5. Last survivor annuity. Annuity makes payments when at least one is alive.
6. Reversionary annuity. Annuity makes payments to second life after first life dies.

A **guaranteed annuity** is one that makes payments for a period regardless of whether the annuitant is alive. Another name for this annuity is a “certain and life annuity”.

Life annuities during their payment period do not have a surrender value. Allowing surrender of such annuities would lead to adverse selection, since individuals who believe they will die soon will surrender their annuities. The pricing of annuities assumes that some people will die earlier, leaving funds to pay annuitants who live longer.

Annuities are not underwritten. Insurance companies make more money if the annuitant dies earlier. It is generally not possible to collect evidence that an annuity purchaser will not live too long.¹³

Disability income insurance **Disability insurance** pays a monthly benefit while the policyholder is disabled. Benefit is a proportion of salary, generally 50–70%, to encourage return to work. Insurance may continue until retirement.

Some time provisions of disability insurance are:

Waiting period , or *elimination period*. Benefits only begin after insured is disabled for this period. Typically 30, 60, 180, or 365 days.

Benefit period . Benefits are payable for at most a certain amount of time. Time is measured from when benefits start, after the waiting period. Typical options are 2 years, 5 years, or to age 65.

Off period . This is the amount of time that must elapse between two disability periods before they are considered separate. This may work both for and against the policyholder; if the off period has not passed, there is no new waiting period, but there is no new benefit period either; the second period’s benefits would only be for the remainder of the benefit period minus the first disability’s period.

Some other characteristics of disability insurance are:

- If the policyholder can do some work but not full-time, the policyholder may be eligible for a lower benefit for partial disability.
- Benefit may be reduced if policyholder receives benefits from other sources, like the government.
- Own job versus any job. Own job insurance pays benefits if the disabled person can’t return to his job, even if other work is available. Any job insurance only pays if the disabled person is so ill that he cannot perform any job. Own job insurance is, of course, more costly.¹⁴
- Employers may purchase disability insurance for employees. Rates are lower than individual policies due to
 1. No adverse selection
 2. Economies of scale
 3. No risk of non-payment of premium
- Some policies have *return to work assistance*. This may pay for retraining or other items that help the disabled person return to work.

¹³But some companies have experimented with offering lower prices for annuities to smokers.

¹⁴Some disability insurance may be “own job” for a period like 5 years, and “any job” after that.

Long term care insurance This insurance pays if someone is ill enough to require a home health aid or a nursing home. Generally a 90-day waiting period. Benefits are triggered when one cannot perform 2 or 3 (depending on the policy) **Activities of Daily Living** (ADLs).

There are 6 ADLs:

1. Bathing
2. Dressing
3. Eating
4. Toileting (ability to go to/from toilet)
5. Continence (bladder/bowel control)
6. Transferring (ability to go from chair to bed to chair; distinguished from toileting)

Severe cognitive impairment may also trigger benefits.

Some other features are:

- Just like disability, there is a benefit period which may be 2–5 years, or, unlike disability, may be for life.
- Just like disability, there is an off period.
- Benefit payments may be on a reimbursement approach subject to a maximum, or may be a fixed amount per month.

Hybrid life insurance/LTC policies may use one of two approaches:

Return of premium approach The excess of premiums over benefit payments is added to the death benefit.¹⁵

Accelerated benefit approach The total LTC benefit cannot exceed the sum insured of the life insurance. The LTC benefits paid are subtracted from the death benefit.

The discussion so far relates to U.S. and Canada. In other countries:

France LTC is popular and cheaper because

1. Trigger for most policies is “severe dependency”, meaning bed- or chair-bound, stricter than “2 ADL” condition.
2. Policies mostly bought as group insurance.
3. Policies bought at younger ages.
4. Lower average benefits.

Payment is fixed annuity.

Germany LTC provided by government social insurance. Additional benefits may be obtained through private LTC insurance. May opt out of government plan (and not pay taxes for it) and use private insurance.

Payment is fixed annuity.

Japan Available stand-alone or with whole life insurance. Benefits may increase with increased dependency.

UK LTC not offered. Instead, there is an *immediate needs annuity* that one purchases with a single premium when moving to a nursing home. Benefits are a fixed annuity paid directly to nursing home. Insurance company may assume higher-than-standard mortality.

Critical illness/chronic illness insurance **Critical illness insurance** pays a lump sum benefit upon a severe condition, such as cancer or heart disease, and then expires.

Chronic illness insurance pays a benefit upon an illness from which one may not recover. Benefit may be lump sum or annuity.

Either of these two may be an **accelerated death benefit rider** to a life insurance contract.

¹⁵I think that for most such policies, there is no other death benefit; the death benefit is just the return of premiums minus benefits.

Forms of ownership of insurance companies

1. **Mutual.** Has no stockholders. Profits are distributed to for-profit (participating) policyholders.
2. **Proprietary**¹⁶. Profits are distributed to shareholders. May have for-profit (participating) policies, and then those policyholders would also get share of profit.

Continuing care retirement communities **Continuing care retirement communities** (CCRCs) are communities for seniors who need varying amounts of assistance. They have three levels of support:

1. Independent living units (ILUs) for residents who do not need extensive help. They may offer housekeeping, transportation to shopping, and similar services.
2. Assisted living units (ALUs), that provide more extensive non-medical help, such as cooking and laundering.
3. A skilled nursing facility (SNF) for those needing medical care. This looks more like a hospital.

Some facilities may also have memory care units (MCU) for residents with dementia.

Funding is done through

1. **Full life care**, with a large entry fee and guaranteed monthly payments only increasing with cost of living adjustments. All costs are covered.
2. **Modified life care**, with a lower entry fee and with monthly payments. Monthly fees increase if the resident moves to the ALU, MCU, or SNF, but these increases are less than the full cost.
3. **Fee for service**. Here payments are made for all services provided at the market rates. Entry fees and monthly payments (at least initially) are lowest.

For full and modified life care contracts:

1. They are insurance contracts in the sense that the CCRC assumes the risk of the costs of higher level care. Residents entering with these contracts must be underwritten and healthy enough to qualify. Residents must be able to initially live in the ILU.
2. Some contracts provide a partial refund upon exit.
3. Some CCRCs offer (partial) ownership of the ILU. The resident purchases the unit and upon death or exit it is sold and the resident gets some of the proceeds.

Average age of entry to CCRC in the U.S. is 80. On the average, full life care residents enter at younger ages than modified life care residents, who enter at younger ages than fee-for-service residents.

Structured settlements When a person is injured, the responsible party must compensate for the loss. Compensation includes paying for medical treatment and lost wages. Rather than paying a lump sum, the insurance company or the court often provides a **structured settlement**. This settlement may include a lump sum for medical payments and other immediate expenses and a life annuity for lost wages, ongoing medical expenses, and other expenses. Providing a structured settlement is better than paying a lump sum because the payment pattern of the structured settlement better matches the pattern of the losses—the lost wages and the ongoing medical expenses resemble an annuity. Thus it relieves the injured party of two risks:

1. The investment risk—the risk of not getting an adequate return on a lump sum
2. The dissipation risk—the risk that the lump sum will be spent too fast

¹⁶Called “stock” in the U.S.

A serious injury may require a whole life annuity, whereas a more minor injury would require a temporary life annuity extending until expected recovery.

Structured settlements are commonly used in Workers Compensation insurance (called workers comp for short), an insurance purchased by an employer to pay benefits to workers injured on the job. They are also commonly used in medical malpractice and motor vehicle accidents.

Usually a structured settlement once determined is final. But after severe injury, there may be a period in which payments are made until the time of maximum mortality improvement, at which point a final settlement is made.

Pensions A pension is a lump sum or a life annuity paid to a retiree by the employer.

The amount of a pension usually varies with the employee's salary and number of years of service.

Pension plans may be defined contribution or defined benefit. In a **defined contribution plan**, the amount set aside each year of employment is specified.

In a **defined benefit plan**, the annual amount paid to the retiree is specified. Typically the annual amount is $n\alpha S$, where n is the number of years of service, α is an accrual rate specified in the plan, and S is the salary base. α is typically 1–2%. The salary base may be

Final salary **The average of salaries in the k years before retirement.**

Career average salary **The average of all salaries earned while in service.**

Career average revalued salaries The average of all salaries with the salaries indexed for inflation.

Defined benefit plans may have **withdrawal benefits**. These are benefits to terminated employees. Typically these would be computed using the $n\alpha S$ formula, and deferred to retirement age.

Defined contribution plans pay a lump sum of the accumulated funds at retirement. They may be used to purchase an annuity.

Old SOA Exam LTAM questions: F18:B4(a), F19:B4(a), S21-1:1,B3(a), F21-A:1, F21-B:1

Multiple choice sample questions: 1.1, 1.2, 1.3, 1.4

Part III

Survival Models